

QuarkyLab

Comprehensive Infrastructure Assurance Review

Evidence-based reliability, security, recoverability, and enterprise-alignment assessment of the NetFRAME / km-cluster home lab

Environment	NetFRAME / km-cluster (7-node Proxmox VE 9.2.3)
Operator	Kyle Mason (machismo0311), Greater Cleveland OH
Review date	2026-07-23
Mode	Read-only. No configuration changes made.
Basis	Live SSH evidence (2026-07-23) + repo audit commit 65f3681
Classification	Internal - contains topology; no secrets reproduced

Prepared as a read-only assurance review. Every finding is tied to environment evidence. No system was restarted, reconfigured, or modified. All change proposals are held for explicit operator approval.

<Home-Lab/docs/audits/quarkylab-comprehensive-review/>

Contents

1	Executive Summary	4
1.1	Overall assessment	4
1.2	Major strengths	4
1.3	Highest risks	4
1.4	Dimension assessments	5
1.5	Top 10 prioritized recommendations	5
1.6	Explicitly accepted risks and where no change is recommended	5
1.7	Confidence and limitations	5
2	Scope and Methodology	5
2.1	Scope	5
2.2	Operating posture: read-only by classification	6
2.3	Evidence approach	6
2.4	Independent validation (multi-method)	6
2.5	What was not done	6
2.6	Deliverable structure	6
3	Evidence Register	7
3.1	Reliability tiers and evidence types	7
3.2	Register	7
3.3	Notable conflicts surfaced	7
4	Infrastructure Inventory	8
4.1	Compute and storage hosts	8
4.2	Network devices	8
4.3	Power devices	9
4.4	Hardware roadmap	9
5	Network Architecture and Correctness	9
5.1	VLAN and Addressing Design	9
5.2	EX3400 Port Map (documented; not live-verified, E-22)	9
5.3	Gateway, DHCP, and DNS	10
5.4	Firewall Policy	10
5.5	Remote Access and Reverse Proxy	10
5.6	Findings	11
6	Proxmox and Virtualization Review	11
6.1	Cluster health	11
6.2	Node consistency	11
6.3	Guest inventory	11
6.4	High availability, replication, and storage definitions	12
6.5	Orphans, stale objects, and discipline	12
7	Storage and Data Integrity	12
7.1	Live pool health	12
7.2	The faulted bulk drive	12
7.3	Controllers and multipath topology	13
7.4	Redundancy posture and aged media	13
7.5	Data integrity and capacity	13
7.6	Recoverability by dataset	13

8 Backup and Disaster Recovery	13
8.1 Backup Posture	13
8.2 3-2-1 Analysis	14
8.3 Recovery Priority and Objectives	14
8.4 Non-Destructive Restore Validation Plan	14
8.5 Per-Domain Recovery Procedures	14
8.6 Recommendations	15
9 Kubernetes, RKE2, and Containers	15
9.1 RKE2 Control Plane	15
9.2 Container Gaps	15
9.3 LXC and Docker Estate	15
10 AI and Jarvis Architecture Review	16
10.1 Architecture and data flow	16
10.2 AI-safety control matrix	16
10.3 Maturity by function	17
10.4 Code versus documentation	17
10.5 Credentials the AI touches	17
11 Monitoring, Logging, and Alerting	17
11.1 Stack and topology	17
11.2 Alert inventory	17
11.3 Coverage	18
11.4 Blind spots and self-dependency	18
11.5 What is done well	19
12 Security Architecture Review	19
12.1 Network segmentation (strong)	19
12.2 Identity and access	19
12.3 Secrets	19
12.4 Unauthenticated services and Kubernetes posture	19
12.5 Reconciliation with the prior security assessment	20
13 Power and UPS Resilience	20
13.1 Live UPS state	20
13.2 Power budget	20
13.3 Scenarios	21
13.4 NUT graceful shutdown and dual-PSU cross-feed	21
13.5 Shutdown and startup sequencing	21
13.6 Efficiency	21
14 Documentation and Operational Readiness	21
14.1 Strengths	21
14.2 Weaknesses	22
14.3 Documentation Gap Register	22
14.4 Recommendation	22
15 Scientific Research and Student Support	22
15.1 Research workloads	23
15.2 Multi-tenant GPU sharing (live-validated 2026-07-02)	23
15.3 Reproducibility and access lifecycle	23
15.4 Readiness assessment	23
16 Enterprise and Fortune 100 Alignment	23
16.1 What Makes This Unusually Strong for a Home Lab	24
16.2 What Would Concern an Enterprise Hiring Manager	24
16.3 Portfolio Framing	25

17 Failure-Mode and Dependency Analysis	25
18 Drift and Contradiction Register	26
18.1 Interpretation	27
19 Risk Register	27
19.1 Summary	27
19.2 High	27
19.3 Medium	28
19.4 Low and Accepted	28
20 Recommendations	28
20.1 By Class	28
20.2 Detailed Treatment of the Top Items	29
20.3 Explicitly No Action	29
20.4 Approval-Driven Next Steps	30
21 Limitations and Confidence	30
21.1 Access and method limitations	30
21.2 Confidence by major conclusion	30
A Command and Activity Log	31
B Asset Register	31
C Service Catalog	32
D Port and VLAN Register	33
D.1 VLANs	33
D.2 EX3400-48P port map	33
E Backup Matrix	33
F Alert Matrix	34
G Dependency Register	35

1 Executive Summary

One-paragraph verdict

QuarkyLab, the NetFRAME / km-cluster home lab, is an unusually mature environment that runs three genuinely production workloads at once: DUNE neutrino-physics GPU research, a multi-tenant student AI/ML teaching cluster, and a self-hosted LLM inference platform. Live read-only verification on 2026-07-23 found the fabric healthy: a 7 of 7 quorate Proxmox cluster, GPUs correctly assigned, backups fresh (14 snapshots dated 2026-07-23), both UPS units online at full charge, and the tracked repositories secrets-clean. The environment is well ahead of typical home-lab practice in operational discipline. The material risks are concentrated, well understood, and mostly already in the operator's own backlog: a single faulted storage drive on a pool that has no second copy, a backup topology that is effectively one-copy / one-site, and two single points of failure (OPNsense and Randy) that the operator is already planning to address.

1.1 Overall assessment

This is a strong environment. The header-line facts were independently confirmed against live hardware rather than taken from documentation (E-01 through E-08). The cluster is quorate with all corosync links connected (E-01); QuarkyLab carries one RTX 8000 (48 GB) and Jarvis two RTX 6000 (24 GB each), which confirms the repository and *contradicts* the review specification's assertion that the assignment was reversed (E-02, E-03, DR-01); the ZFS datastore pool is clean and last night's backups completed (E-06, E-08). Nothing found during this review indicates an imminent outage. The weaknesses are real but bounded, and the operator has already produced a self-audit (commit 65f3681, 2026-07-22) that names most of them.

1.2 Major strengths

- **Deterministic AI-operations safety.** The netframe-monitor / Jarvis stack is genuinely enterprise-grade in its safety design: a single deterministic evidence engine that every report path uses, model self-confidence stripped and forbidden, a three-action human-gated remediation allowlist with no autonomous execution, universal policy screening including the chat and Discord surfaces, untrusted-telemetry handling, and full survivability if the model is offline (E-14).
- **Disciplined runbook and disaster-recovery culture.** Formal RCAs and AARs, ADRs, an offline break-glass credential path independent of the cluster, a node-unreachable triage runbook, OPNsense serial-console DR with age-encrypted config backups, and a laptop-recovery runbook (E-21).
- **Real, applied network segmentation.** Seven VLANs, BMCs isolated to VLAN 20 with an *applied* egress clamp, a VLAN 30 to VLAN 1 block, and DNS pinholes for isolated VLANs (E-21). This is deployed, not aspirational.
- **Config-as-code monitoring.** Grafana alerting, exporters, and textfile collectors are versioned; there are dead-man heartbeats for the pipeline itself (E-15).
- **Honest self-audit already in flight.** The operator is mid-way through a reconciliation branch that catalogs the documentation drift found here.

1.3 Highest risks

1. **HIGH Faulted bulk drive on a pool with no second copy (QL-R01 + QL-R02).** mpathv is FAULTED in raidz2-2, which now holds only one of two parity drives; a second failure in that vdev loses the pool, and bulk (which holds research and media data) is neither a PBS source nor target (E-04). Replacement is pending.
2. **HIGH Backups are effectively one-copy / one-site (QL-R03).** Every backup lives only on Randy: one PBS, one ZFS pool, one rack, one UPS bus (E-08).
3. **HIGH OPNsense is a single point of failure (QL-R04).** One VM on pve2 provides routing, firewall, DHCP, and inter-VLAN gateway for the whole LAN.
4. **HIGH Randy is a single storage node (QL-R05).** PBS, research NFS, and RKE2 persistent volumes all depend on it.
5. **HIGH Critical-alert delivery is unverified (QL-R11).** The degraded pool was surfaced by a backup-verify staleness alert, not by the direct ZfsPoolDegraded rule; whether that critical rule actually reached Discord could not be confirmed read-only.
6. **MEDIUM A live GitHub token sits in plaintext at rest (QL-R07).** ~/github_helper.py contains a classic PAT; it is gitignored and never committed, but usable if the workstation is compromised (E-17). Two other plaintext secret files sit alongside it.

7. **MEDIUM Documentation drift (QL-R19).** Recent changes landed in CLAUDE.md but not in README, topology, or the docs mirror; two review-spec "corrections" are inverted versus live reality and must not be applied.

1.4 Dimension assessments

- **Reliability:** Strong at the compute/cluster layer (7/7 quorum, pinned GPU stacks intact); weakened by concentrated SPOFs (OPNsense, Randy) and HA fencing left in standby.
- **Security:** Well above home-lab norm - applied segmentation, least- privilege API tokens, working SIEM. Residual: at-rest workstation secrets, unauthenticated LAN AI/registry endpoints, guest SIEM blind spot.
- **Recoverability:** Good documentation and an offline break-glass path; limited by the single backup site and one unverified restore-readiness gap (pve1 unbacked).
- **Power resilience:** Both UPS online and healthy, but UPS-A is loaded to 59 percent with roughly 9.5 minutes of runtime (E-07), an imbalance worth rebalancing.
- **AI-system reliability:** Excellent safety posture; the remaining lift is calibration/eval, not trustworthiness.
- **Scientific and student workloads:** Genuinely well engineered - SLURM shard plus per-job MPS VRAM caps plus apptainer isolation, research preemption, and reproducible container images.
- **Enterprise alignment:** Comparable-to-mature in monitoring, docs, IaC, and secrets discipline; material gaps in HA/redundancy and offsite DR.

1.5 Top 10 prioritized recommendations

1. Replace the faulted bulk drive and scrub (QL-REC01, urgent).
 2. Verify the ZfsPoolDegraded alert actually reached Discord (QL-REC02, read-only).
 3. Add a second backup copy / restore the 3-2-1 principle (QL-REC03).
 4. Back up the standalone pve1 guests (QL-REC04).
 5. Rotate the at-rest GitHub PAT and the two plaintext secret files (QL-REC05).
 6. Deploy prometheus-pve-exporter for cluster/quorum/guest metrics (QL-REC06).
 7. Add the missing alert rules: PBS-stale, verify-failed, broad cert-expiry (QL-REC07).
 8. Rebalance UPS loads and confirm NUT graceful-shutdown sequencing (QL-REC08).
 9. Reconcile EX3400 SSH access and capture switch config to git via Oxidized (QL-REC09).
 10. Land the documentation reconciliation PR (D-01 through D-10) (QL-REC11).
- Building the OPNsense CARP HA pair (QL-REC10) is the highest-value medium-term structural improvement and removes the top availability SPOF.

1.6 Explicitly accepted risks and where no change is recommended

The following are sound decisions that should stand: the pinned 6.14.11-9-pve kernels on the GPU nodes must not be upgraded (QL-REC20); the raidz1 vdevs on QuarkyLab and Jarvis are an accepted single-parity tradeoff given hot spares, clean scrubs, and low utilization (QL-R21); single-node observability is a reasonable posture at this scale given the dead-man heartbeats (QL-R10); and the VLAN 1 egress lockdown remains deliberately unenforced pending its prerequisites. Stable systems should not be changed without a compelling reason, and most of this environment is stable.

1.7 Confidence and limitations

Confidence in the live-verified conclusions (cluster health, GPU assignment, storage state, backup freshness, power state) is **high**: each was confirmed by direct read-only query and cross-checked against the prior audit. Confidence in the network-control-plane conclusions is **medium**: OPNsense was not queried (no credentials used, by design, E-23) and the EX3400 could not be inspected over SSH (key authentication is broken, E-22 / R-09), so switch and firewall specifics rest on repository documentation rather than live output. This was a strictly read-only review; nothing was restarted, reconfigured, or modified, and no credential was used. See Section 21 for the full limitations register.

2 Scope and Methodology

2.1 Scope

This review evaluated the QuarkyLab / NetFRAME home lab across 21 assessment dimensions: reliability, availability, recoverability, security, network correctness, configuration consistency, monitoring quality, alert quality, storage integrity, backup

quality, power resilience, energy efficiency, capacity planning, operational maturity, documentation quality, automation safety, AI-system reliability, scientific-workload support, student-environment isolation, recruiter and hiring-manager presentation, and Fortune-100 enterprise alignment.

In scope were only systems, repositories, and services that belong to the home lab: the 7-node km-cluster plus the standalone pve1 host, Randy and the DS4246 shelf, the Juniper and UniFi switching, OPNsense, Headscale, the RKE2 cluster, the Docker/LXC service estate, the Ollama/Open WebUI/llm_router AI stack, the Grafana and Wazuh observability planes, the UPS/PDU/power layer, and the operator's GitHub repositories. No unrelated external system was probed. External services already used by the environment (package repositories, GitHub, container registries, certificate authorities, DNS resolvers) were treated as passive dependencies, not targets.

2.2 Operating posture: read-only by classification

Every command was classified before execution as read-only/low-risk, read-only/resource-intensive, configuration-changing, service-affecting, destructive, or external-target interaction. Only read-only/low-risk commands were run without approval. No configuration-changing, service-affecting, destructive, or external-interaction command was executed. Concretely, this review performed no restarts, reboots, reloads, migrations, deletions, rotations, or deployments; used no discovered credential; injected no failures; and did not enforce the VLAN 1 egress lockdown, in line with the standing decision that its prerequisites are not yet met. Where confidence that an action was non-disruptive fell below the required threshold, the action was not taken and the limitation was recorded (see Section 21).

2.3 Evidence approach

The review compared five state layers where possible: declared (the review specification and prior intent), configured (files and definitions), deployed (running processes and guests), observed (live telemetry), and documented (the repository). Documentation was not treated as true merely because it exists, and live output was not treated as authoritative where it represents only transient runtime state.

A key input was the operator's own prior audit (commit 65f3681, 2026-07-22), which had already captured per-node live evidence and built a canonical fact ledger. That corpus is one day old and was used as generated evidence to be re-verified, not trusted blindly. This review independently re-ran the highest-stakes checks against current live hardware.

2.4 Independent validation (multi-method)

For every high-severity conclusion, at least two independent read-only methods were used. Examples actually applied in this review:

- **GPU assignment:** live `nvidia-smi` on both R730s (E-02, E-03) versus the review specification and the fact ledger. The live reading settled the contradiction in favor of the repository (DR-01).
- **Storage health:** live `zpool status` on Randy (E-04) versus the ledger's canonical geometry and the prior audit's finding on `mpathv`.
- **Cluster and guests:** live `pvecm status` plus `pvesh get /cluster/resources` (E-01, E-06) versus the documented node and service placement, which surfaced the Grafana/Headscale relocation drift.
- **Power:** live `upsc` readings (E-07) versus the reliability assessment's UPS-imbalance finding, which confirmed UPS-A is the loaded unit.
- **Backups:** live PBS snapshot listing dated 2026-07-23 (E-08) versus the documented backup schedule.

2.5 What was not done

No failure injection, no restore into production, no restarts, no configuration changes, no credential use, and no live query of OPNsense (E-23) or the EX3400 (E-22). These boundaries are honored throughout; findings that depend on them are marked medium-confidence and carry a precise follow-up request.

2.6 Deliverable structure

The deliverables are this LaTeX report (sections plus appendices), a set of machine-readable registers under `data/` (evidence, assets, services, drift, risk, recommendations, backup matrix, alert matrix), and text-based topology diagrams under `diagrams/` (physical, logical, service-dependency, recovery-dependency, power). The `evidence/` directory describes evidence locations and collection methods rather than copying sensitive output.

3 Evidence Register

Every finding in this review traces to a numbered evidence item. The register below records the source, the system it came from, the exact command or path, its age, a reliability tier, and the limitations that bound how far it can be trusted. The machine-readable form is `data/evidence-register.csv`.

3.1 Reliability tiers and evidence types

Evidence is typed as *live* (a direct read-only query executed during this review, 2026-07-23), *live-captured* (the prior audit's node captures, one day old), *generated* (assessment reports produced by prior tooling), *inferred* (conclusions read out of source code or configuration), or *manual* (operator-maintained documentation). Live evidence is weighted highest for current-state claims; documented evidence is weighted highest for intent and history. Where the two conflict, the conflict itself is recorded rather than silently resolved: the most consequential examples are the GPU assignment (live overrides the specification, DR-01), the OPNsense version (documented as 25.1.12, not re-verified live, E-23 / DR-02), and Randy's PVE version, where the live reading of 9.2.3 supersedes the documented 9.1.1 (E-04 / DR-10).

3.2 Register

ID	Source / system	Path or command	Age / rel.	Type / limitation
E-01	Live SSH / pve3	pvecm status; corosync-cfgtool -s	live / high	live; transient runtime state
E-02	Live SSH / quarkylab	nvidia-smi; zpool list; qm list; sinfo	live / high	live; confirms GPU vs stale spec
E-03	Live SSH / jarvis	nvidia-smi; zpool list; systemctl is-active	live / high	live; confirms 2x RTX 6000
E-04	Live SSH / randy	zpool status bulk/datastore; pbs datastore list	live / high	live; multipath -ll returned empty
E-05	Live SSH / pve1	pveversion; pct list; apt list --upgradable	live / high	live; 142 pending vs audit 133
E-06	Live SSH / pve3	pvesh get /cluster/resources; storage.cfg; ha-manager	live / high	live; cluster-wide snapshot
E-07	Live SSH / pve3	upsc triplite; upsc midatlantic	live / high	live; instantaneous load only
E-08	Live SSH / randy	ls -dt /datastore/{ct,vm,host}/*/2026-07-23	live / high	live; filesystem listing
E-09	Repo audit / Home-Lab	audit/findings/*.md (commit 65f3681)	1 day / high	generated; two spec corrections inverted
E-10	Repo audit / Home-Lab	audit/live/*.txt	1 day / high	live-captured; superseded where re-run
E-11	Repo audit / Home-Lab	audit/facts/ledger.yaml	1 day / high	inferred; canonical-vs-stale table
E-12	Repo / Home-Lab	CLAUDE.md	2 days / high	manual; most-accurate doc
E-13	Repo / Home-Lab	README.md; topology/; vault MOC	1 day / med	manual; lags CLAUDE.md
E-14	Code read / netframe-monitor	*.py; constitution/; eval/	live / high	inferred; static read not runtime trace
E-15	Code read / monitoring-stack	grafana-provisioning/alerting/*.yaml	1 day / high	inferred; rules not eval-traced
E-16	Code read / netframe-dashboard	serve.py; README.md	1 day / high	inferred; some sources fall back to mock
E-17	Secrets sweep / home dir	grep + git ls-files across repos	live / high	live; PAT at rest in github_helper.py
E-18	Prior assessment / enterprise	report/*.tex; risk-register.tex	10 days / med	generated; some items since resolved
E-19	Prior assessment / security	report/*.tex; MASTER-TODO.md	live / high	generated; reconciled 2026-07-22
E-20	Prior assessment / reliability	report/*.tex	9 days / med	generated; windowed items pending
E-21	Repo docs / Home-Lab	vault/Runbook/*.md (AARs, DR, CARP)	live / high	manual; operator narrative
E-22	Live SSH / ex3400	ssh ex3400 → DENIED	live / high	access-limit; confirms R-09
E-23	Not collected / opnsense	no API creds used; VM 100 untouched	n/a	not-collected; 25.1.12 per AAR only
E-24	Repo / ansible playbooks	roles/{hardening,backup_verify}	1 day / high	manual; --check only, not applied

3.3 Notable conflicts surfaced

The register makes three conflicts explicit. First, the review specification's GPU "correction" is inverted versus live hardware and must not be applied (E-02 / E-03). Second, the OPNsense version is documented as 25.1.12 but was not re-verified live; roughly a dozen references still say 25.7, which is the future CARP target, not the running version (E-23). Third, Randy reports PVE 9.2.3 live while the documentation still says 9.1.1, a drift the prior audit did not catch (E-04, DR-10).

4 Infrastructure Inventory

Every entry below is confirmed from live read-only SSH evidence collected 2026-07-23 (E-01 through E-08) unless explicitly flagged as documented-only. Where the live reading disagrees with a repository document, the live value is treated as canonical and the discrepancy is carried into the drift register (Section 18).

4.1 Compute and storage hosts

The estate is a seven-node Proxmox VE cluster (km-cluster) plus one standalone Proxmox host (pve1). The two Dell R730 servers and the SuperMicro storage node carry the production research, inference, and storage workloads; the four HP EliteDesk nodes carry the network edge and core services.

Host	Role	CPU / RAM	GPU	PVE	Kernel / status
pve1	Standalone Pi-hole / homepage host (NOT a cluster member)	Mac Mini, 16 GB	none	9.1.9	7.0.0-3-pve; 142 pending updates; openipmi failed (no BMC)
pve2	OPNsense host + cluster node	i7-8700, 32 GB	none	9.2.3	7.0.12-1-pve; reboot-required; excluded from hardening role
pve3	Core services + rke2-cp1	i7-8700, 48 GB	none	9.2.3	7.0.12-1-pve; e1000e TSO/GSO-off mitigation persisted
pve4	Grafana stack + rke2-cp2	i5-7500T, 32 GB	none	9.2.3	7.0.12-1-pve; sustained load ~5
pve5	Headscale, Pi-hole2, rke2-cp3, HAOS	i5-7500T, 32 GB	none	9.2.3	7.0.12-1-pve
QuarkyLab	ML / DUNE research / student GPU	2× E5-2699 v4 (44c/88t), 512 GB	1× RTX 8000 48 GB	9.2.3	6.14.11-9-pve (pinned); kernel + full nvidia-* held
Jarvis	LLM inference	2× E5-2687W v4, 384 GB	2× RTX 6000 24 GB	9.2.3	6.14.11-9-pve (pinned); kernel held only (QL-R12)
Randy	Storage / PBS / Jellyfin / rke2 storage worker	2× E5-2690 v3 (24c/48t), 128 GB ECC	RX (transcode)	580 9.2.3	7.0.12-1-pve; docs say 9.1.1 (DR-10)

Live confirmations of contested facts

- **GPU assignment (DR-01):** nvidia-smi confirms QuarkyLab carries the single RTX 8000 (46080 MiB) and Jarvis carries two RTX 6000 (24576 MiB each), driver 550.163.01. The review specification's "correction" inverting these is wrong and must not be applied.
- **Kernel pins intact:** both GPU nodes run the pinned 6.14.11-9-pve; the pin protects the NVIDIA 550 stack. Never upgrade the kernel or change GRUB_DEFAULT on either node (QL-REC20).
- **Randy PVE version (DR-10):** live pveversion reports pve-manager/9.2.3, whereas CLAUDE.md, README, and the fact ledger still state 9.1.1. New drift found by this review.
- **Patch lag:** pending-update counts range 63 (Randy) to 142 (pve1); all cluster nodes report reboot-required (QL-R13).

4.2 Network devices

Device	Model	Mgmt IP	Role / status
EX3400-48P	Juniper EX3400	192.168.10.50	Core switch, JunOS 23.4R2-S7.4, RSTP root (priority 4096), dual PSU. SSH key auth broken (QL-R16). xe-0/2/3 DOWN - SFP EEPROM speed mismatch (DR-14).
USW-24-250W UDR	UniFi UniFi Router Dream	via UDR 192.168.10.2	Access switch, trunk to ge-0/0/46. VLAN 1 wireless controller / AP - NOT the WAN edge.
EX2300-48P AP7901	Juniper EX2300 APC	-	Lab / CCNA access switch. Managed PDU on ge-0/0/38.

The EX3400 could not be inspected live: SSH refused the offered key (E-22), confirming the open reconciliation item R-09. Switch state is therefore taken from repository runbooks and the SNMP feed the dashboard already scrapes, not from a live console pull.

4.3 Power devices

Two UPS units protect the estate (live readings in Section 13): Middle Atlantic UPS-OL2200R (UPS-B, servers and shelf) and Tripp Lite SMART1500VA (UPS-A, switch, UniFi, EliteDesk nodes). Both are online at full charge.

4.4 Hardware roadmap

Two hardware changes are in flight and should be tracked as capacity/redundancy items: Randy RAM expansion 128 GB → 192 GB (+64 GB on hand), and a Dell R730xd being commissioned as a second PBS/backup target and Jellyfin transcode node. The R730xd directly addresses the single-backup-node gap (QL-R03, QL-REC03) and its power impact is estimated in Section 13.

5 Network Architecture and Correctness

This section reconstructs the network from repository evidence and live cross-checks rather than trusting any single diagram. Two constraints bound its confidence: the EX3400 core switch could not be inspected live because key-based SSH is broken (E-22, tracked as R-09 and QL-R16), and OPNsense was deliberately not touched because no read-only API credential was used for this review (E-23). Items that rest only on repository documentation, not on a live reading, are marked *inferred*. See diagrams/logical-topology.mmd and diagrams/physical-topology.mmd.

5.1 VLAN and Addressing Design

The estate runs seven VLANs on the EX3400 (ELS), gatewayed by OPNsense. The design is a clean default-deny segmentation model with a dedicated out-of-band plane.

ID	Name	Subnet	Purpose / population
1	mgmt/default	192.168.10.0/24	Proxmox hosts, corosync, services, switch mgmt (dense)
20	trusted/OOB	192.168.20.0/24	iDRAC x2 + IPMI (BMCs) + Ares jump leg
30	servers	192.168.30.0/24	NFS/PBS/GPU egress; dual-homed GPU and storage nodes
40	IoT	192.168.40.0/24	smart-home, sensors (sparse)
50	VoIP	192.168.50.0/24	CP-8841 phones, FreePBX (deferred)
60	guest	192.168.60.0/24	isolated guest WiFi
70	lab	192.168.70.0/24	CCNA/experimental; standalone sandbox node

The GPU and storage nodes (QuarkyLab, Jarvis, Randy) are dual-homed: management and corosync stay on VLAN 1 while NFS, PBS, and internet egress ride VLAN 30. Corosync was deliberately kept off VLAN 30 to hold the cluster ring on stable L2, which is sound practice.

5.2 EX3400 Port Map (documented; not live-verified, E-22)

Port	Attachment	Mode	VLANs
ge-0/0/22	Jarvis nic2 (onboard 1G)	Trunk	native 1 (+ tagged 30, now vestigial)
ge-0/0/24	QuarkyLab nic2	Trunk	native 1 + tagged 30
ge-0/0/30, /32, /44	iDRAC/IPMI OOB	Access	20 only
ge-0/0/38	APC AP7901 PDU	Access	1
ge-0/0/41	Ares mgmt/jump	Dual	native 1 + tagged 20
ge-0/0/45	EX2300 uplink	Trunk	all
ge-0/0/46	UniFi USW-24 Port 24	Trunk	native 1 + tagged 20/30/40/50/60/70

Port	Attachment	Mode	VLANs
xe-0/2/0	Randy nic3 (10G ConnectX-3)	Trunk	1 + 30 (storage/NFS)
xe-0/2/2	Jarvis enp132s0 (10G ConnectX)	Access	30 (dedicated storage/egress)
xe-0/2/3	UniFi SFP2 DAC	MEDIUM DOWN EEPROM 10G/1G speed mismatch (DR-14)	

The switch is RSTP root (bridge-priority 4096) with dual PSUs. One residual hardware fault: xe-0/2/3 is down due to a DAC EEPROM speed mismatch, costing a redundant UniFi path; the documented fix is a fiber SFP+ at both ends (DR-14). A load-bearing ELS gotcha is recorded repeatedly: native-vlan-id must be set at the physical-interface level, not under unit 0 family ethernet-switching. A stale note in two documents still claims the EX3400 does not support it at all; that is false and would cause an operator to build a broken trunk (drift DR-08, doc fix D-08).

5.3 Gateway, DHCP, and DNS

OPNsense (VM 100 on pve2) is the single LAN router, firewall, DHCP server, and inter-VLAN gateway, live-confirmed running with high egress counters (E-06). DHCP moved to the Kea backend after an empty-Kea outage on 2026-07-20; Kea holds 12 reservations and serves all seven VLAN scopes. WAN is dual-path with single NAT: WAN1 is a Spectrum public block terminated directly on vtnet0 (avoiding double-NAT), WAN2 is a FirstNet 5G hotspot on vtnet2 as an automatic failover gateway group.

DNS is genuinely highly available: primary Pi-hole .177 (pve1 CT103) and secondary .178 (pve5 CT108, a nebula-sync full mirror on a 15-minute timer). OPNsense DHCP hands out both resolvers on every scope, so clients fail over automatically. The resolver chain is client to Pi-hole to Unbound on .1 to internet. A residual concern is that both Pi-holes forward to a single Unbound instance (enterprise finding R-03), which has historically been the root of intermittent DNS behavior. Critical containers (.177 Pi-hole, .181 NPM, .183 Grafana, .148 Homepage) were made static-in-container so a DHCP-server outage cannot drop routing, a good hardening move.

5.4 Firewall Policy

The inter-VLAN posture is default-deny with explicit allows, and several clamps have been applied and verified:

- **BMC isolation and egress clamp (verified 2026-07-17):** all three BMCs sit on VLAN 20; only Ares reaches them; a floating quick-block prevents VLAN 20 to internet, so compromised BMC firmware cannot reach VLAN 1, VLAN 30, or the internet.
- **VLAN 30 to VLAN 1 block:** one surgical rule blocks servers to 192.168.10.0/24, with a narrow allow for Prometheus scrape on :9100.
- **DNS pinholes:** isolated VLANs (IoT/VoIP/Guest/Lab) keep :53 to the Pi-hole alias while remaining isolated.
- **Deny logging:** all eight VLAN deny rules log to Wazuh/Loki for lateral-movement visibility.

Standing decision respected - VLAN 1 egress lockdown is NOT enforced. The management-plane to internet limit is documented as deliberately deferred, and this review does not recommend enforcing it without first satisfying every prerequisite: UniFi discovery and required multicast/broadcast behavior, legitimate network-device outbound connectivity, RKE2 and container-registry/CDN dependencies, Mist or other vendor callbacks, administrative access paths, DHCP reservations, an emergency access path, current firewall API permissions, and a tested rollback. Enforcing it prematurely is a credible way to cause a self-inflicted outage.

5.5 Remote Access and Reverse Proxy

Headscale (self-hosted, LXC 105 on pve5, v0.29.1) is the internal admin mesh only, with nine nodes; students and researchers do not use it. Student and researcher ingress is a Cloudflare Tunnel (quarkylab.kylemason.org) with no inbound ports and the server IP hidden. Public web services are fronted by Nginx Proxy Manager (LXC 101, admin :81 firewalled to Ares only, F-05):

Hostname	Backend	TLS / auth
vault.kylemason.org	Vaultwarden .30.182	CF DNS-01 / login

Hostname	Backend	TLS / auth
grafana.kylemason.org	Grafana .183 (pve4)	CF DNS-01 / Grafana
homepage.kylemason.org	Homepage .148	CF DNS-01 / basic auth
wazuh.kylemason.org	Wazuh .184	CF DNS-01 / Wazuh
chat.netframe.local	Open WebUI .30.185	internal / admin acct
llm.netframe.local	Jarvis llm_router .31:8000	HTTP, Pi-hole-scoped DNS

5.6 Findings

- **HIGH** OPNsense is a single point of failure for all routing, DHCP, and inter-VLAN traffic (QL-R04). A CARP HA plan exists (QL-REC10).
- **MEDIUM** No automated network-configuration capture, and EX3400 SSH is broken (F-N1, R-08/R-09, QL-R16). Reconcile SSH and deploy Oxidized to snapshot switch and firewall config to Git nightly (QL-REC09).
- **MEDIUM** Single Unbound recursion point on .1 (enterprise R-03).
- **LOW** xe-0/2/3 down from an SFP EEPROM mismatch (DR-14); a residual Amazon Echo/Fire tablet remains on flat VLAN 1 pending IoT cutover.
- **INFO** IPv6 is not configured on the VLANs and is default-denied; no explicit jumbo/MTU tuning is documented on the 10G fabric.

6 Proxmox and Virtualization Review

6.1 Cluster health

The cluster is healthy and fully quorate. Live `pvecm status` from pve3 (E-01) reports seven nodes, seven votes, Quorate: Yes, expected votes 7, quorum 4, config version 13, ring ID 1.9eda7. `corosync-cfgtool -s` shows every peer connected on the single knet link over VLAN 1. Node IDs map to pve2 (.204), pve5 (.203), pve4 (.202), pve3 (.201), QuarkyLab (.179), Jarvis (.31), and Randy (.187). pve1 (.193) is correctly absent - it is a standalone host, not a cluster member.

6.2 Node consistency

All seven cluster members run PVE 9.2.3. The four EliteDesk nodes and Randy run kernel 7.0.12-1-pve; the two GPU nodes run the pinned 6.14.11-9-pve. The standalone pve1 lags at PVE 9.1.9 / kernel 7.0.0-3-pve. The only version anomaly inside the cluster is Randy's package set reporting 9.2.3 while documentation still records 9.1.1 (DR-10); functionally the node is healthy.

6.3 Guest inventory

Cluster-wide guest state from `pvesh get /cluster/resources` (E-06):

ID	Name	Node	Purpose	onboot	Backup
CT101	nginx-proxy	pve3	Reverse proxy (NPM)	1	randy-pbs
CT102	vaultwarden	pve3	Secrets manager	1	randy-pbs
CT103	grafana	pve4	Metrics/logs/alerting stack	1	randy-pbs
CT105	headscale	pve5	Tailnet coordinator	1	randy-pbs
CT106	homepage	pve3	Dashboard	1	randy-pbs
CT107	openwebui	pve3	Chat UI to llm_router	1	randy-pbs
CT108	netframe-pihole2	pve5	Secondary DNS	1	randy-pbs
VM100	OPNsense	pve2	Router / firewall / DHCP	1	randy-pbs
VM104	wazuh	QuarkyLab	SIEM	1	randy-pbs-10g
VM110	homeassistant	pve5	Home automation (HAOS)	1	randy-pbs
VM201	rke2-cp1	pve3	Kubernetes control plane	1	randy-pbs
VM202	rke2-cp2	pve4	Kubernetes control plane	1	randy-pbs
VM203	rke2-cp3	pve5	Kubernetes control plane	1	randy-pbs

Every production guest has onboot=1 and a nightly backup job. The cluster-side counterpart to CT103 Grafana and CT105 Headscale confirms both moved to pve4 and pve5 respectively (the 2026-07-16 de-concentration), contradicting README/topology which still draw them on pve3 (DR-03, DR-04).

6.4 High availability, replication, and storage definitions

ha-manager status reports quorum OK but fencing standby (CRM watchdog standby) with no HA groups defined (E-06). A node failure therefore does not auto-migrate guests; recovery is manual (QL-R18) - a defensible choice on consumer EliteDesk hardware, but it should be an explicit, documented decision. pvesr status shows no replication jobs.

Two PBS storage targets are defined in /etc/pve/storage.cfg: randy-pbs (server 192.168.10.187, keep-all=1) carries the LXCs and VM100/VM110/rke2; randy-pbs-10g (server 192.168.30.187 over the 10G VLAN 30 path, keep-daily=7, keep-weekly=4, restricted to nodes QuarkyLab/Jarvis/Randy) carries the Wazuh VM. Both point at the same datastore and fingerprint. Enterprise repositories are disabled and the no-subscription repo is enabled on cluster nodes.

6.5 Orphans, stale objects, and discipline

The only orphan of note is CT104 on pve1 (old homepage), which documentation records as retired 2026-06-24 but which is still running (DR-11, R-12). No orphaned disks, stale snapshots, or abandoned templates were surfaced by the live captures.

Strength: kernel-pin discipline

The GPU nodes hold their kernel via GRUB_DEFAULT and (on QuarkyLab) the full nvidia-* package set, deliberately protecting a validated driver stack from an incidental apt upgrade. This is mature change control. The one gap is that Jarvis holds only the kernel, not the nvidia-* packages - it should mirror QuarkyLab's fuller hold set (QL-R12, QL-REC12).

7 Storage and Data Integrity

7.1 Live pool health

Five ZFS pools were inspected live (E-02, E-03, E-04). Four are healthy; one carries a faulted device.

Pool	Host	Size	Geometry	State
datastore	Randy	36.7 T	4× RAIDZ2 (three 6-wide + one 4-wide Seagate)	ONLINE, 0 errors
bulk	Randy	80.0 T	3× RAIDZ2 (8+8+6-wide)	DEGRADED - mpathv FAULTED
workspace	QuarkyLab	10.9 T	6-wide RAIDZ1 + 1 hot spare (PERC H330 JBOD)	ONLINE, 0 errors
tank	Jarvis	9.09 T	5-wide RAIDZ1 (HBA330 IT)	ONLINE, 0 errors
scratch	Jarvis	186 G	single SSD	ONLINE, 0 errors

The live datastore layout confirms the disputed documentation fact D-04: the fourth vdev (raidz2-3) is four Seagate drives by-WWN, in-pool, with no unallocated spares. The bulk geometry confirms D-05: three vdevs totalling 80 T, superseding the stale "2×8-wide / 58.2 T" still in topology and Proxmox Cluster.md.

7.2 The faulted bulk drive

CRITICAL **QL-R01 - top operational priority.** zpool status bulk (E-04) shows device mpathv in vdev raidz2-2 as FAULTED with 3 read / 14 write errors and "too many errors". The device is a SEAGATE ST4000NM0023 (4 TB SAS), WWN 5000c500631a54fb; both SAS paths are dead, so this is a failed physical drive, not a multipath flap. It corresponds to the DS4246 slot-15 drive that failed 2026-07-21 (incident NF-INC-2026-07-22).

Key mitigating facts, all live-verified: the pool reports errors: No known data errors; RAIDZ2 tolerates one drive loss per vdev, so raidz2-2 still has one parity drive of protection remaining; the last scrub (Sun Jul 19) completed with zero errors; and two DS4246 bays are free for an immediate zpool replace. The exposure is the window until replacement: a second failure in raidz2-2 before the resilver completes would lose the bulk pool.

Why this is severe despite being tolerated

The bulk pool holds media, archive, and - critically - researcher fernanda's DUNE data, and bulk is neither a PBS source nor a PBS target (QL-R02). It has no second copy anywhere. A degraded single-parity vdev protecting data that has no backup is the pairing that elevates this from routine to urgent. Recommended action QL-REC01: replace mpathv, scrub, and confirm Scrutiny actually flagged the failing drive.

7.3 Controllers and multipath topology

Randy uses two distinct controllers by design: an AVAGO 3108 MegaRAID for the boot mirror (relocated 2026-07-01 after its original slot died - do not reuse the dead slot) and an LSI 9207-8e in IT mode for the DS4246 data path. The DS4246 is dual-SAS cabled with multipath, so every shelf drive presents as an mpathX device. The single enclosure is itself a single point of failure for bulk (H-09, QL-R05 context): loss of the shelf or its cabling takes the whole pool offline.

7.4 Redundancy posture and aged media

The two GPU-node pools are RAIDZ1 (QL-R21): workspace tolerates one loss but carries a hot spare; tank tolerates one loss with no spare; scratch is a single SSD with no redundancy (scratch by intent). Several disks are aged SAS. RAIDZ1 on old disks carries resilver-window risk - a second fault during rebuild loses the pool. This is an accepted risk today given clean scrubs and low utilisation, but future rebuilds should prefer RAIDZ2 and keep spares on hand.

7.5 Data integrity and capacity

Scrubs are scheduled and recent (datastore and bulk both scrubbed Jul 19, zero repaired). Scrutiny monitors roughly 79 drives across Randy, QuarkyLab, and Jarvis; the one open verification item is confirming it surfaced mpathv rather than staying silent (paired with QL-R11 on alert delivery). Utilisation is low across the board - datastore 176 G of 36.7 T, bulk 80.7 G of 80 T, workspace 144 G of 10.9 T - so there is no near-term capacity pressure; the risk is redundancy and backup, not space.

7.6 Recoverability by dataset

Guest images and configs are recoverable from PBS within roughly 24 h RPO (Section 8). The exceptions are the two datasets with no independent copy: bulk (including research data) and pve1's CT103/CT104. Ares' own restic backup, notably, targets sftp:randy:/mnt/bulk/backups/ares - it lands on the degraded bulk pool, so the workstation backup shares the same at-risk failure domain and should be repointed once a second target exists.

8 Backup and Disaster Recovery

Backup execution is healthy and current; backup *architecture* is the weak point. Live evidence (E-08) shows 14 snapshots dated 2026-07-23 from the previous night's runs, with the PBS datastore at 97 GB used of 23 TB. Every cluster guest is protected daily. The problem is that all of those copies live on one node.

8.1 Backup Posture

Asset	Target	Schedule / re-tention	Notes
LXC 101/102/106/107	randy-pbs .10.187	02:00 7d/4w daily,	NPM, Vaultwarden, Homepage, Open WebUI
LXC 103 Grafana	randy-pbs	02:00 daily	observability stack
LXC 105 Headscale, 108 pihole2	randy-pbs	02:00 daily	pihole2 mirrors CT103 config
VM 100 OPNsense	randy-pbs + config repo	03:00 03:17 daily +	config also age-encrypted to GitHub
VM 104 Wazuh	randy-pbs-10g .30.187	03:00 daily	10G path

Asset	Target	Schedule / re-tention	Notes
VM 110 HomeAssistant	randy-pbs	03:30 daily	HAOS
VM 201/202/203 RKE2	randy-pbs	04:00 daily	etcd not separately backed up
QuarkyLab workspace	PBS host backup	05:30 daily	fernanda refreservation floor
pve1 CT103/CT104	HIGH NONE	never	primary Pi-hole + old homepage unbacked (QL-R06)
bulk pool data	HIGH NONE	never	media + fernanda research, no second copy (QL-R02)
Ares /home	restic to	daily 04:03	encrypted, but lands on the DEGRADED bulk pool
Ares ~/.claude	randy:/mnt/bulk/... randy:/datastore/ ares-claude	hourly	memory/transcripts, secrets excluded

8.2 3-2-1 Analysis

The estate is effectively 1-1-1: one PBS instance, on one ZFS pool, in one rack, on one UPS bus (QL-R03, **HIGH**). Three specific exposures compound it:

- **pve1 guests are unprotected** (QL-R06 / R-03): the primary Pi-hole CT103 and the retired CT104 have no vzdump job. This is partly mitigated because the secondary Pi-hole CT108 is backed up and nebula-syncs the primary's config, so DNS state is rebuildable.
- **The bulk pool has no second copy** (QL-R02): it is neither a PBS source nor a target, so the DUNE research data and media it holds exist only on the currently degraded DS4246 shelf.
- **The Ares restic repository lands on bulk** (randy:/mnt/bulk/ backups/ares), the same degraded pool, so the workstation backup shares the shelf's fault domain.
Encryption is partial: the OPNsense config backup is age-encrypted and the Ares restic repository is encrypted, but the PBS datastore itself is not encrypted at rest.

Do not assume a green job is a recoverable backup

PBS reports success and runs a weekly verify plus garbage collection, which is good, but no full restore has been rehearsed for most guests. A backup that has never been restored is an untested backup.

8.3 Recovery Priority and Objectives

Recovery priority, highest first: (1) power and core network, (2) OPNsense routing and DHCP, (3) DNS, (4) Randy storage and PBS, (5) Vaultwarden and certificates, (6) the app and AI tiers, (7) research and student services. Estimated objectives from the backup matrix: RPO is 24 h for guest images (daily vzdump) and 1 h for Ares memory; RTO ranges from roughly 10-30 minutes per guest restore to several hours for a full cluster rebuild. The OPNsense config RPO is effectively 24 h with a 15-30 minute restore.

8.4 Non-Destructive Restore Validation Plan

No restore was performed during this review. The recommended validation, all non-destructive and to be scheduled with operator approval, is: (a) run a PBS verify job and confirm zero failures; (b) restore one representative guest to a scratch VMID on a node with free space, boot it isolated, confirm it starts, then destroy the scratch guest; (c) test-decrypt the OPNsense age config and the break-glass snapshot. Never restore into a production VMID.

8.5 Per-Domain Recovery Procedures

The estate has an unusually strong recovery-documentation culture; the procedures below already exist and were reviewed:

- **OPNsense:** serial console via `qm terminal 100` from pve2 plus a DR-tested age-encrypted `config.xml` restore; SSH is disabled by design so console is the only shell.
- **DNS:** `pct start 103` on pve1, or fail to the secondary .178; `pihole restartdns` if the container is up but DNS is stuck.

- **Cluster/secrets loss:** the break-glass age snapshot held on both Ares and Randy breaks the Vaultwarden-in-cluster circular dependency (QL-R08), letting an operator read credentials with Vaultwarden, NPM, and the cluster all down.
- **Network device:** the EX3400 config is not automatically captured, so switch rebuild is slow and manual (gap, QL-REC09).
- **RKE2:** rebuild rests on etcd quorum; etcd is not separately snapshotted beyond VM-level backup.
- **Git and docs:** GitHub plus local clones; the repos are the source of truth for config-as-code.

8.6 Recommendations

Add a second, independent backup target (QL-REC03): the planned R730xd running its own PBS with PBS-to-PBS remote sync, plus an offsite copy to Backblaze B2 for the datastore and bulk/fernanda. Add the pve1 vzdump job (QL-REC04). See `diagrams/recovery-dependencies.mmd`.

9 Kubernetes, RKE2, and Containers

9.1 RKE2 Control Plane

RKE2 runs a three-node HA control plane on VMs 201/202/203 (pve3/pve4/pve5, .51- .53), all live-confirmed running (E-06). The API is fronted by a kube-vip VIP at 192.168.10.54:6443; the CNI is Cilium and load balancing is MetalLB over the .71- .75 pool. etcd runs as an embedded quorum across the three control-plane nodes. The cluster is intentionally CPU-only: GPU scheduling is deferred because SLURM and Ollama own the cards, and running RKE2 GPU scheduling on the same single card would contend for exclusive device access.

Randy joined as a bare-metal storage worker with kubelet reservations sized to measured hardware and a `node.netframe.io/role=storage:NoSchedule` taint, so only tolerating infrastructure DaemonSets land there and ZFS/PBS are not cgroup-throttled. Persistent storage is an `nfs-client` StorageClass backed by Randy datastore/k8s. A documented operational detail worth preserving: agents register on the supervisor port :9345, not the apiserver :6443.

9.2 Container Gaps

- **MEDIUM** No NetworkPolicies: the pod network is flat with no default-deny (QL-R15 / K8S-02).
- **MEDIUM** No Pod Security Admission (K8S-03), and Kubernetes secrets are likely unencrypted at rest (K8S-04).
- **MEDIUM** The private registry is anonymous and a single replica (QL-R14 / K8S-01); its TLS is provided by step-ca, and a RegistryCertExpiringSoon alert exists.
- **LOW** Image pinning and floating latest tags were not exhaustively audited; pinning discipline should be confirmed as workloads grow.
- **INFO** etcd is protected only at the VM-backup level; a dedicated etcd snapshot schedule would speed control-plane recovery.

The recommended treatment (QL-REC15) is to add default-deny NetworkPolicies, Pod Security Admission, secrets encryption, and registry authentication before the cluster carries meaningful workloads. None of this should be applied without a maintenance window and rollback.

9.3 LXC and Docker Estate

The operator convention is one service per LXC and "no Docker unless explicitly requested," which keeps blast radius small and is a genuine strength. The running guests were enumerated live (E-06): NPM (CT101), Vaultwarden (CT102), Grafana stack (CT103 on pve4), Headscale (CT105), Homepage (CT106), Open WebUI (CT107), secondary Pi-hole (CT108), plus VMs for OPNsense, Wazuh, Home Assistant, and the three RKE2 control-plane nodes. Docker appears only inside a few LXCs where it is warranted: the Grafana stack (Grafana, Prometheus, Loki, InfluxDB, Scrutiny) in CT103, Vaultwarden in CT102, and the Homepage dashboard in CT106.

All guests are set `onboot=1`. Compose-versus-deployed drift could not be fully verified read-only, since that would require exec into each container; it should be spot-checked during normal maintenance. Consistent with the read-only mandate, this review did not pull images, recreate containers, or restart any workload.

The one duplicate to resolve is CT104 on pve1, an old homepage instance that documentation says was retired on 2026-06-24 but that is still running (DR-11 / R-12). Confirm it is the retired skeleton, then stop and disable it (QL-REC17).

10 AI and Jarvis Architecture Review

This section audits the netframe-monitor / Jarvis AI-operations system as a production operational system, not as a demo. The central question is whether a language model is ever permitted to state a conclusion as fact, take an action, or become a command without a deterministic gate and explicit human approval. The short answer, verified against the code (E-14), is no: the design is safety-first, and it is one of the signature strengths of this lab.

10.1 Architecture and data flow

The system separates a deterministic plane from a model plane, and the deterministic plane is load-bearing for every safety-critical function.

- **Collection (no model).** netframe_monitor.py runs every 15 minutes via netframe-run.sh, SSHing read-only probes across seven nodes with a dedicated monitor_key and writing last_run.json plus an append-only history.jsonl. It includes end-to-end user-impact probes (LLM_ROUTER through NPM, and LLM_ROUTER_CONFORMANCE reporting config, runtime, and firewall as three independent dimensions that are never collapsed).
- **Report paths.** netframe_interpret.py (now), netframe_daily.py, netframe_predict.py (deterministic least-squares forecast), netframe_monthly.py, and netframe_chief.py (weekly synthesis). Every path imports both netframe_policy and netframe_evidence and ends with the same sequence: model prose, then a deterministic evidence section, then netframe_policy.enforce().
- **Admission screen.** netframe_admission.py vetoes resource-consuming probes on 72B residency, GPU-busy, interactive user, maintenance window, or minimum interval, and fails closed when GPU state is unknown, so the monitor never becomes the workload it monitors.
- **Deterministic alerter.** netframe_alert.py fires node-down Discord messages straight from last_run.json on UNREACHABLE transitions, independent of Grafana, using fixed templates (nothing for a model to influence).
- **Config drift.** netframe_confdrift.py fingerprints interfaces, ssh, systemctl, and env over the read-only path and diffs against a human-blessed baseline.
- **Interfaces.** netframe_console.py serves a read-only HTTP API with no execute endpoint; both the console and report pages are iptables-locked to localhost plus NPM.
- **Model routing.** llm_router.py defaults to local Ollama qwen2.5:72b, serves a rag model over a numpy cosine index, and escalates to claude-opus-4-8 on an explicit flag, a claude- model, or local failure.
- **Audit.** netframe_audit.py maintains a hash-chained, tamper-evident, append-only ledger mirrored to Loki at write time.

10.2 AI-safety control matrix

Control	Implementation (evidence E-14)	Verdict
Single shared evidence engine	netframe_evidence.py is called by all six paths; the Findings contract forbids the model stating a confidence; strip_model_confidence removes any residual confidence text on the free-form console path	Strong
Provenance and evidence age	Every factor carries provenance; freshness() with FRESHNESS_MAX_S = 5400s caps confidence at 40 and renders STALE; unverifiable citations contribute zero and trigger a 30 ceiling	Strong
Deterministic confidence	evidence_quality and confidence are pure-stdlib floors-plus-ceilings over telemetry; the module states "no model is consulted here"; tightest ceiling wins	Strong
Human-gated remediation	netframe_remediate.py: 3-item Tier-1 allowlist only, list-form argv (no shell), requires explicit approve plus confidence >= 95; anything else refuses and prints a manual workflow	Strong
Universal policy screening	netframe_policy.enforce() is the single gate on every model-to-operator path, including chat; the Discord alerter is deterministic templates with nothing to bypass	Strong
Model output never a command	_guard_actions strips shell commands and code fences from chat output and honors only an allowlisted [action:id] tag; the console, not the model, renders the gated command	Strong
Untrusted telemetry	Telemetry wrapped in BEGIN/END UNTRUSTED markers with a data-not-instructions prompt, plus a deterministic injection tripwire (honest caveat: a tripwire for overt imperatives, not a boundary)	Good
Model-offline survivability	Collector, alerter, confdrift, admission, forecast tables, and evidence scoring are all no-model; node-down alerts fire before any LLM step	Strong
Failed model calls	Every path degrades to a deterministic fallback; policy.enforce fails LOUD but not closed (emits an UNSCREENED stamp rather than suppressing)	Strong

10.3 Maturity by function

Function	Rating	Basis
Monitoring assistant	Strong / production	Deterministic 15-min collector, dead-man heartbeat, Grafana-independent alerter, locked read-only surfaces
Incident analysis	Good	Deterministic blast-radius from a knowledge graph, two-axis evidence and confidence per finding
Capacity planning	Solid (deterministic)	Least-squares slope and days-to-90%; math never delegated to the model
Documentation	Strong	Auto now/daily/weekly/monthly reports, hash-chained ledger, unified memory dashboard
Change planning	Recommend-only	Chief P0-P3 ranking; propose emits a manual workflow; confdrift human-bless gate
Remediation	Minimal / human-gated	3-action allowlist, confidence gate, never autonomous (constitution authority_limits.md)

10.4 Code versus documentation

The `jarvis-ai-ops` LaTeX corpus is explicitly propose-and-assess. The code is now ahead of those assessments on the two biggest historical gaps: the two-axis evidence scoring proposed in NF-AIOPS-005 is fully implemented, and the tamper-evident audit ledger flagged as open in JAR-12 is built. Two residual gaps remain and are reflected in the recommendations:

- **Remediation approval gate (JAR-08, QL-REC13).** Report and chat confidence are now deterministic, but the Tier-1 execution gate still reads a caller-supplied confidence integer rather than the `netframe_evidence` engine. The gate is safe by construction (allowlist plus explicit approval) but the number is not yet wired to computed evidence.
- **Config conformance.** `confdrift` compares production to a blessed snapshot of production, not to Git. The specific `llm_router` env-drift incident class is now covered by the separate conformance probe, but a general conform-to-Git target is not yet built.

Assessment

As a home-lab AI-operations platform this is unusually mature. The trust foundation - single deterministic evidence engine, universal policy screen, human-gated allowlist remediation, untrusted-input handling, and full model-offline survivability - is enterprise-grade. Critical alerting does not depend on any single AI model, and no model output can become an action or a shell command without a deterministic gate and explicit approval. The remaining lift is the intelligence layer (evaluation harness, retrieval calibration, Git conformance), not the safety layer. This directly satisfies the review requirement that deterministic monitoring remain available if the AI is offline and that remediation cannot occur without approval.

10.5 Credentials the AI touches

Reported by type and location only, values redacted (E-14). `ANTHROPIC_API_KEY` in `/etc/llm_router.env` (0600, unset means local-only); `DISCORD_BOT_TOKEN` and the operator user id in `/etc/netframe-alert.env` (root 0600); the read-only `monitor_key` used by the collector, `confdrift`, and the scoped `wazuh-restart` wrapper; and the `restic` backup passphrase file. The `restic` passphrase is the one single-point risk: if it lives only on Jarvis, loss of that host makes the encrypted backup unreadable, so it must also live in Vaultwarden. No hardcoded secrets were found in any source file read.

11 Monitoring, Logging, and Alerting

11.1 Stack and topology

Observability is a Grafana, Prometheus, Loki, InfluxDB, and Scrutiny stack running together in CT103 on pve4 (192.168.10.183), live-confirmed (E-06). Alerting uses Grafana Unified Alerting straight to Discord with no Alertmanager. Two contact points exist: an infrastructure channel (`discord-alerts`) and a UPS channel (`discord-ups`). Prometheus was reported at 15 of 15 targets up (E-15), which means the silent not-scraping failure mode is clean. Config lives as code in a private repository, which is a genuine operational-maturity strength.

11.2 Alert inventory

The following rules were extracted from the provisioning repository (E-15). GAP- rows are missing alerts, not deployed ones.

Alert	Source	Sev	Threshold	Disposition
ClusterNotQuorate	corosync textfile	crit	quorate < 1, 2m	Keep
ClusterVotesBelowExpected	corosync textfile	warn	votes deficit > 0, 5m	Keep
InstanceDown	node/blackbox	crit	up < 1, 2m	Keep
PiholePrimaryDown	blackbox DNS	crit	probe .177 < 1, 2m	Keep
PiholeSecondaryDown	blackbox DNS	warn	probe .178 < 1, 2m	Keep
ZfsPoolDegraded	zpool textfile	crit	health > 0, 5m	Verify delivery (R-02)
GpuTempHigh	nvidia-smi textfile	warn	temp > 87C, 5m	Keep
GpuMemoryHigh	nvidia-smi textfile	warn	VRAM used > 90%, 10m	Keep
DiskAlmostFull	node_filesystem	warn	root > 90%, 10m	Extend to pools
LowMemory	node_memory	warn	avail < 8%, 10m	Keep
RegistryCertExpiringSoon	blackbox tls	warn	cert < 12h, 15m	Broaden to all certs
BackupVerifyReportStale	Randy textfile	warn	age > 26h, 15m	Keep (caught mpathv)
BackupVerifyFailing	Randy textfile	warn	pass < 1, 5m	Consider critical
AresBackupStale	Randy textfile	warn	age > 26h, 15m	Keep
NetframeMonitorStale	Jarvis heartbeat	warn	age > 1h, noData alerting	Keep (dead-man)
UPS battery low	PeaNUT	crit	charge < 50, 2m	Keep
UPS runtime low	PeaNUT	crit	runtime < 300s, 1m	Keep
PrimaryWANDown	pve2 textfile	crit	gateway < 1, 1m	Keep
FirewallBlockRateSpike	Loki filterlog	warn	> 500 / 5m	Keep
NetworkDeviceConfigChange	Loki syslog	warn	commit > 0	Keep
NetworkSyslogSilent	Loki stream	warn	< 50 / 15m, noData alerting	Keep (dead-man)
GAP-quorum-guest	(needs pve-exporter)	-	per-guest down	Missing (QL-REC06/07)
GAP-pbs-verify-stale	-	-	PBS verify failed/stale	Missing (QL-REC07)
GAP-host-auth-failure	-	-	repeated auth failures	Missing (DETECT-01c)
GAP-cert-expiry-broad	-	-	any internal/LE cert near expiry	Missing (only registry to-day)
GAP-2nd-channel	-	-	alert egress independent of WAN/Discord	Missing (QL-REC14)

11.3 Coverage

Signal	Status	Notes
Availability	Covered	InstanceDown, Pi-hole probes, PrimaryWANDown, quorum
Latency	Partial	OPNsense per-WAN latency on dashboard only; DNS probe is up/down
Errors	Partial	Journal errors as a dashboard chip; FirewallBlockRateSpike; no app error-rate rule
Saturation	Covered	LowMemory, GpuMemoryHigh; no CPU-saturation rule
Capacity	Partial	DiskAlmostFull on root only; pool/non-root capacity not alerted
Temperature	Covered	GpuTempHigh; no CPU or ambient temp alert
Power	Covered	UPS battery and runtime via PeaNUT
Storage health	Covered	ZfsPoolDegraded plus Scrutiny SMART; delivery of the critical rule unverified
Cert expiry	Narrow	Registry TLS only; other certs not monitored
Backup success	Covered	BackupVerify stale/failing, AresBackupStale
Auth failures	Gap	No host auth-failure alert rule (dashboard chip only)
Config drift	Partial	Hardening-drift chip and NetworkDeviceConfigChange; no host config-drift alert

11.4 Blind spots and self-dependency

- **Single alert egress (QL-R09, DETECT-01d).** All alerts leave over one Discord webhook across the WAN. A WAN or Discord outage blinds alerting, and the deterministic node-down message shares that path. A second out-of-band channel is recommended (QL-REC14).
- **Co-located stack (QL-R10).** Metrics, logs, and alert evaluation all run in one CT on one node. Losing that node loses monitoring and stops alert evaluation.
- **Loki unauthenticated.** auth_enabled: false on VLAN 1 makes logs readable and poisonable by any VLAN 1 host.
- **SIEM scope (QL-R22).** Wazuh agents run only on the eight hypervisors; guests, OPNsense, the switch, and endpoints are blind to the SIEM.

The silent-alert lesson

The degraded bulk pool was masked for roughly 71 hours. The backup-verify job ran as an Ares user cron; the laptop suspends overnight with no catch-up, so three nightly runs were silently dropped and backup_verify_* froze. The condition was ultimately surfaced by BackupVerifyReportStale, not by ZfsPoolDegraded. Whether

ZfsPoolDegraded actually fired to Discord is unverified read-only (QL-R11, R-02) **HIGH**. A silent critical is worse than no monitoring; verifying that delivery path is the single highest-value monitoring action. The scheduler itself has since been moved to a persistent systemd user timer.

11.5 What is done well

Dead-man heartbeats already exist (NetframeMonitorStale, NetworkSyslogSilent, both alerting on noData), a corosync quorum textfile alert has been added, WAN-down detection is wired from OPNsense dpinger, and the entire configuration is version-controlled. The recommended next steps are a read-only prometheus-pve-exporter for guest and quorum metrics (QL-REC06), the backup-stale, verify-failed, and broad cert-expiry rules (QL-REC07), and the second alert channel (QL-REC14).

12 Security Architecture Review

This is a defensive review only. No credential was used, no exploit was attempted, and all secret values are redacted to type, path, and a redacted indicator.

12.1 Network segmentation (strong)

Segmentation is the standout security control. Seven VLANs are defined and live on the EX3400 (E-21, architecture digest). Both server-grade BMCs and the Randy IPMI were moved off flat VLAN 1 to VLAN 20 with factory credentials rotated, and an egress clamp (applied and verified 2026-07-17) prevents a compromised BMC from reaching VLAN 1, VLAN 30, or the internet. A single surgical rule blocks VLAN 30 to VLAN 1 except a narrow Prometheus scrape. IoT, VoIP, Guest, and Lab are isolated with a DNS pinhole so they keep ad-blocking without lateral reach. All eight deny rules log to Loki and Wazuh. There is no inbound port exposure; external reach is via NPM with TLS and via a Cloudflare Tunnel with no open ports. The VLAN 1 egress lockdown is deliberately deferred as disproportionate gold-plating, which this review respects and does not recommend enforcing without the documented prerequisites.

12.2 Identity and access

Administration uses shared root over key-only SSH. The Ansible hardening role installs a strong sshd policy (key-only, PermitRootLogin prohibit-password, no password or keyboard-interactive) but has been run in check mode only and is not yet enforced on the nodes (DR-12), so the fleet is not yet hardened in production. There is no MFA on administrative interfaces, and the prior assessment flagged weak Ares admin credentials (enterprise R-23). These are reasonable for a single-operator lab but are the items an enterprise reviewer would raise first.

12.3 Secrets

The committed-secret posture is clean: the Home-Lab full-history gitleaks scan is clean across 300-plus commits, and every tracked hit is a placeholder or shell-variable reference (E-17). This review adds three findings the committed-secret audit could not see, all at rest on the Ares workstation and none ever committed to git:

Type	Path	Redacted	Status
GitHub PAT (classic)	~/github_helper.py:4	ghp_...	Untracked, gitignored, never in history; live token at rest
Generic API key	~/apikey.txt	key=...	Untracked; real value on disk
Wazuh API password	~/wazuh-api-newpw.txt	(0600)	Untracked; plaintext at rest

The GitHub PAT (QL-R07) should be rotated and moved to an environment or credential store, and the two files removed once their values are relocated (QL-REC05). Separately, six repositories lack a .gitignore and the tracked .bash_aliases and .ssh/config publish internal topology in a public-intent superproject (QL-REC18).

12.4 Unauthenticated services and Kubernetes posture

Ollama and llm_router accept unauthenticated LAN requests but are iptables-locked to known clients; the private registry is anonymous (K8S-01) and Loki is unauthenticated (QL-R14). RKE2 runs with a flat pod network, no NetworkPolicies, no Pod

Security Admission, and likely unencrypted secrets (QL-R15, K8S-02/03/04). These are lateral-movement value if a LAN host is compromised rather than direct external exposure. TLS is a strength: grade-A internal certificates from step-ca and Let's Encrypt via Cloudflare DNS-01. Patch lag is tracked separately (QL-R13).

12.5 Reconciliation with the prior security assessment

ID	Sev	Finding	Status
F-001	High	Untrusted IoT/printer/mobile on mgmt VLAN	Open (needs UDR admin)
F-003	High	Workstation cleartext FTP / EOL SMB/NFS	Open
F-006	High	Unencrypted cluster-root SSH keys	Open
K8S-01	High	Anonymous private registry	Open
NET-01	Med	OOB/BMC asymmetric isolation	Done (VLAN 20 egress clamp)
VIRT-01	Med	Hypervisor firewall disabled	Partial
VIRT-02	Med	Mgmt API on all interfaces	Partial (5/7)
HOST-01	Med	Password root SSH on some hosts	Jarvis hardened; others via role (DR-12)
AI-01	Med	Unauthenticated LLM endpoint	Mitigated (iptables lock)
DETECT-01b/c/d	Info	Guest SIEM, adversary alerts, 2nd channel	Open

Positive controls

No inbound exposure; L3 segmentation with default-deny and deny-logging; applied BMC egress clamp; authenticated cluster heartbeat; unprivileged containers; least-privilege API tokens for the dashboard and monitor; grade-A TLS; a working SIEM with dead-man alerts; an age-encrypted break-glass credential path independent of the cluster. For a single-operator home lab this is a strong, coherent posture. The material gaps are enforcement of the hardening role, rotation of the at-rest workstation credentials, and detection coverage of guests and the network edge.

13 Power and UPS Resilience

13.1 Live UPS state

Both UPS units were read live via NUT (upsc, E-07):

UPS	Status	Battery	Load	Runtime / feeds
Middle Atlantic UPS-OL2200R (UPS-B)	OL	100%	36%	~29 min; R730s, Randy, DS4246
Tripp Lite SMART1500VA (UPS-A)	OL	100%	59%	~9.5 min; EX3400, UniFi, EliteDesk nodes

Both are online at full charge with clean input voltage (113-114 V). The concern is the imbalance: UPS-A (the network and core-services plane) sits near 60% load with under ten minutes of runtime, while UPS-B (the heavy servers) sits at 36% with nearly half an hour. This confirms the prior reliability finding (P-03 / R-18) and is logged as QL-R17.

13.2 Power budget

Estimated from live GPU draw (E-02/E-03) plus platform baselines. GPUs were near-idle at capture: QuarkyLab RTX 8000 at 3.7 W, Jarvis RTX 6000 pair at ~62 W and ~43 W.

Load group	Idle (est.)	Peak (est.)	UPS
QuarkyLab R730 (idle GPU)	~220 W	~650 W (GPU + CPU)	B
Jarvis R730 (2× RTX 6000)	~300 W	~700 W (72B inference)	B
Randy + DS4246 shelf	~300 W	~450 W	B
4× EliteDesk nodes	~120 W	~260 W	A
EX3400 + UniFi + PDU	~120 W	~180 W	A
Estimated total	~1060 W	~2240 W	A+B

These are engineering estimates, not metered values; only the GPU draw and UPS load percentages are measured. Adding the planned R730xd backup node would add roughly 150-350 W depending on transcode activity, and should land on UPS-B (or better, cross-fed) - but UPS-B has the headroom for it whereas UPS-A does not.

13.3 Scenarios

Normal load: both UPS comfortably within capacity. **Peak (GPU + inference):** UPS-B rises toward 55-60% and its runtime falls proportionally, still adequate for a graceful shutdown. **Extended outage:** UPS-A is the binding constraint - at ~9.5 min it must trigger a graceful shutdown of the network and core-services plane quickly, and DNS/routing will drop before the servers do. Rebalancing load onto UPS-B (QL-REC08) is the single highest- value power action.

13.4 NUT graceful shutdown and dual-PSU cross-feed

NUT monitors both UPS units (confirmed live via `upsc -l` on pve3). The open audit item R-07 remains: confirm `upsmon` runs on every node with a correct `SHUTDOWNCMD` and that staged `HOSTSYNC`/timers sequence a clean shutdown (storage last, edge first in the reverse of the recovery order). Equally important and not yet evidenced: the dual-PSU R730s should have one cord on each UPS. If both cords are on UPS-B, the redundant PSU buys nothing against a UPS-B fault. This should be physically verified.

13.5 Shutdown and startup sequencing

The safe shutdown order is the inverse of the recovery dependency graph (Section 17, diagram `recovery-dependencies.mmd`): shed non-critical guests, then app tier, then AI/research, then RKE2, then core services, then OPNsense, then storage (Randy last so backups and NFS stay available longest), then hypervisors. Startup reverses this: power and network, hypervisors, quorum, Randy storage, OPNsense, DNS, then up the stack. This sequencing should be captured as a one-page runbook; today it is implicit.

13.6 Efficiency

Two efficiency wins are already banked from the prior reliability work: CP-001 (GPU persistence / P0 idle fix on Jarvis, roughly -72 W) and CP-002 (CPU governors on pve3/4/5/Randy, roughly -90 W total). Remaining low-risk candidates: spinning down or powering off the DS4246 while bulk is near-empty (P-04, roughly -100 W - but weigh against the faulted-drive replacement work), and correcting the UPS-A/UPS-B imbalance to keep both units in their efficient load band. Consistent with the review's intent, none of these should be pursued at the expense of reliability; the drive replacement and load rebalance come first.

14 Documentation and Operational Readiness

The operative question for this phase is whether a competent administrator who has never seen NetFRAME could operate and recover it from the written record alone. The short answer is: for day-to-day operation and most single-service recovery, yes; for a full cold rebuild, mostly, provided they know that CLAUDE.md and the recent vault AARs supersede the front-facing documents.

14.1 Strengths

The documentation culture here is well above the home-lab norm and is a genuine portfolio asset.

- CLAUDE.md is the single most accurate document in the estate and is consistently ahead of every other source (E-12). It carries the authoritative node, IP, service, storage, and safety-note tables and reads as a living operations context file.
- A disciplined change record: roughly sixty dated runbooks and AARs under `vault/Runbook/`, ten Architecture Decision Records with context, decision, consequences, and alternatives, and formal post-incident RCAs (pve3 e1000e NIC hang, DHCP/Kea outage, backup-verify scheduler, Randy PCIe slot).
- Real disaster-recovery material: OPNsense serial-console DR with an age-encrypted config repository, a break-glass credential snapshot on both Ares and Randy that does not depend on the cluster, a node-unreachable triage guide, and a full laptop-recovery runbook (E-21).
- The Obsidian vault at `Home-Lab/vault/` is declared the authoritative knowledge base and is structured into ADR, Compute, Infrastructure, Networking, and Runbook areas with a master map of content.
- Documentation-as-code with CI enforcement: a GitHub Actions job regenerates the topology Mermaid from `topology/inventory.yml` and fails the build if the drawing drifts from source.

14.2 Weaknesses

The dominant defect is **MEDIUM** one-directional drift (QL-R19): recent changes land in CLAUDE.md and the AARs but do not propagate outward to README.md, topology/, vault/00 - Homelab MOC.md, and the docs/ mirror.

- Stale clusters D-01 through D-10: Grafana still drawn on pve3 (moved to pve4), Headscale still on pve3 (moved to pve5), Randy CPU still v4/28c (corrected to v3/24c/48t), the DS4246 bulk pool still 58.2 T/2-vdev (expanded to 80 T/3-vdev), OPNsense still 25.7 in about twelve places (live is 25.1.12; 25.7 is the future CARP target), homepage still shown on pve1 CT104 (now pve3 CT106), and the EX3400 native-vlan-id note still wrong.
- Duplicate sources of truth: the same runbook lives in three homes (runbooks/, docs/, and vault/Runbook/) with real content drift in the Randy commissioning runbook and cosmetic drift elsewhere.
- Historical documents presented in the present tense: runbooks/EX3400-SSH-Auth-Failure-RCA.md (references the switch at .10.2, now the UniFi Dream Router) and homelab-setup.md need clear "historical/superseded" banners (D-10).

14.3 Documentation Gap Register

State key: **OK** well documented and current, **DRIFT** documented but stale in secondary sources, **THIN** exists but incomplete, **TRIBAL** lives mainly in the operator's head or chat history.

Category	State	Recommended canonical source
Architecture / topology	DRIFT	topology/ generated from inventory.yml; reconcile with CLAUDE.md
Physical inventory / asset	THIN	Asset register (this review, Appendix B) + CLAUDE.md tables
Rack layout	THIN	vault/Rack Layout.md (add rack SVG, open loop)
IP address management (IPAM)	OK	CLAUDE.md network tables; consider a dedicated IPAM note
VLAN / trunk design	DRIFT	vault/Networking/ + EX3400 config once captured (QL-REC09)
Firewall policy	OK	vault/Runbook/Security-VLAN-Segmentation-*
DNS / DHCP	OK	CLAUDE.md safety notes + DNS-HA runbook
Proxmox / virtualization	OK	CLAUDE.md + Cluster-Update-Procedure
Storage	DRIFT	CLAUDE.md Storage section (docs mirror lags)
Backup	OK	backup-verify role + this review Appendix E
Kubernetes / RKE2	OK	RKE2 phase runbooks
AI services / Jarvis	OK	netframe-monitor/constitution/ + jarvis-ai-ops .tex
Monitoring / alerting	OK	netframe-monitoring-stack repo (config-as-code)
UPS / power	THIN	vault/Power Distribution.md + this review Section 13
Startup / shutdown sequence	THIN	Formalize from recovery-dependencies diagram + NUT plan
Incident response	OK	AAR corpus
Disaster recovery	OK	OPNsense DR, break-glass, laptop-recovery
User access	OK	QuarkyLab account onboarding runbook
Student onboarding / offboarding	THIN	add-cluster-key.sh; offboarding automation planned
Scientific workload deploy	THIN	Researcher guide (.tex) + SLURM phase docs
Change management	OK	Manual-deploy policy + CI/CD runbook
Secret rotation	TRIBAL	Vaultwarden is the store; rotation procedure undocumented
Certificate renewal	THIN	step-ca; only registry cert monitored (QL-REC07)
Firmware / lifecycle	THIN	Per-node BIOS/iDRAC notes scattered; consolidate
Hardware replacement	OK	Randy PCIe slot + storage expansion runbooks

14.4 Recommendation

Adopt a single-source generation model for the docs/ tree so that the vault (or CLAUDE.md) is the sole authority and docs/ and runbooks/ are generated or reduced to pointers (QL-REC11). The immediate action is to land the D-01 through D-10 documentation reconciliation, which is already staged on the audit/2026-07-22-reconciliation branch, flowing corrections from CLAUDE.md outward and keeping the 25.7 reference only in the CARP HA plan. Secret-rotation and certificate-renewal procedures are the two categories most reliant on tribal knowledge and should be written down next.

15 Scientific Research and Student Support

QuarkyLab is a genuinely multi-tenant platform: it carries a production neutrino-physics ML workload, a shared teaching environment for computer-science students, and supporting AI tooling, all on one RTX 8000. This section separates what works as a home-lab convenience from what would be required for a formally governed student and research service.

15.1 Research workloads

The physicist fernanda runs a production DUNE reconstruction ML workload on the RTX 8000 (live-confirmed present and idle at review time, E-02). A DUNE Agent RAG pipeline over the dunereco codebase assists onboarding, and a second researcher builds a separate RAG pipeline (ARID) under a scoped exemption. Research data is protected from student starvation by a refreservation=2T floor on the researcher dataset.

15.2 Multi-tenant GPU sharing (live-validated 2026-07-02)

One physical Turing card (no MIG) is shared safely between research and roughly 15 to 20 students per semester:

- **Scheduling.** SLURM gres/shard with 8 shards allows up to 8 concurrent GPU jobs while leaving the card in Default compute mode so the researcher runs native.
- **Hard VRAM cap.** A per-job NVIDIA MPS instance inside each student container enforces a per-job VRAM limit (about 6 GB per shard), chosen over SLURM gres/mps to avoid forcing exclusive-process mode on the researcher.
- **Isolation.** job_submit.lua wraps every student job in aptainer exec --nv --network=none with a private home, read-only shared data, and a RAM bound; cgroup.conf ConstrainDevices=yes denies /dev/nvidia* to any job without a GPU GRES.
- **Priority.** The research partition (PriorityTier=100) preempts student jobs (PreemptMode=REQUEUE, not suspend, so freed VRAM is actually released), and a queued job auto-restarts.
- **Fairness.** Multifactor fairshare with a 3-day decay favors students who have used the GPU least; studentqos caps one running job per student; students are batch-only.

The Phase 04 validation table shows every case passing: no-GRES jobs denied the device, a student request for the whole card rejected, VRAM out-of-memory at the expected cap, 3 then 8 concurrent students, the researcher native at 47.8 GB, and preemption reclaiming the card.

15.3 Reproducibility and access lifecycle

Reproducibility rests on a curated Aptainer image (base.sif, pinned torch and TensorFlow) run from a local copy and refreshed daily so a storage hiccup cannot block a launch. Access is key-only SSH over a Cloudflare Tunnel with no inbound ports; onboarding is a scripted add-cluster-key.sh that validates the key and creates a locked-down account, and a group sshd policy denies password auth, all forwarding, and sudo. Offboarding removes the tailnet node and the user account.

15.4 Readiness assessment

Strength (production-grade)	Gap for a governed service
Real device isolation via cgroups + aptainer + MPS cap	No per-student network isolation within the node
Research preemption and fairshare	Research and student data both single-copy (QL-R02)
Per-user ZFS quotas and read-only shared data	No formal data-retention or privacy policy
Reproducible pinned container images	Offboarding not yet automated (planned)
Batch-only, no interactive un-capped access	Student containers not covered by Wazuh (QL-R22)

Verdict

For a home lab this is exceptional: the GPU multi-tenancy is designed and validated to a standard most small HPC groups would accept, and the research floor plus preemption model is a real answer to the hardest problem in shared GPU teaching. The gaps are governance rather than mechanism: a second copy of research data (QL-R02, QL-REC03), a written retention and privacy posture, automated account lifecycle, and extending SIEM coverage to student workloads would move this from a well-run lab to a defensible institutional service.

16 Enterprise and Fortune 100 Alignment

This comparison is deliberately about capability and operational discipline, not budget. The lab is not penalized for lacking commercial products where an open-source practice delivers the same control. Each dimension is classified as **Comparable** to mature enterprise practice, **Partially aligned**, **Developing**, **Material gap**, or **N/A at this scale**.

Dimension	Classification	Evidence and realistic improvement
Asset management	Partially aligned	CLAUDE.md tables + this review's register; no live CMDB. Improve: keep the asset register generated.
Configuration management	Partially aligned	Ansible hardening + backup_verify roles, lint-clean; but hardening is --check only, not enforced (DR-12). Improve: apply and gate drift.
Network segmentation	Comparable	Seven VLANs, default-deny inter-VLAN, applied BMC egress clamp, VLAN30 to VLAN1 clamp, deny-logging. A genuine strength.
Identity management	Developing	Shared root SSH, no MFA, single operator. Improve: per-admin accounts, MFA on Vaultwarden/Grafana.
Secrets management	Partially aligned	Vaultwarden + offline break-glass age snapshot. Gap: plaintext creds at rest on Ares (QL-R07), no rotation procedure.
Monitoring	Partially aligned	Prometheus/Grafana/Loki, 15/15 targets, dead-man heartbeats. Gaps: single-node stack (QL-R10), missing PVE/cert rules (QL-REC06/07).
Logging	Partially aligned	Loki + OPNsense/EX3400 syslog + Wazuh on hosts. Gaps: Loki unauthenticated, guests not in SIEM (QL-R22).
Incident management	Comparable	Formal RCAs and AARs with rollbacks; better than many small enterprises.
Change management	Partially aligned	Manual-deploy-only policy, CI lint/branch-protection, pre-push hooks. Gap: no formal change calendar/approvals beyond self.
Backup	Partially aligned	PBS nightly, verified, fresh (E-08). Gap: 3-2-1 not met (QL-R03), pve1 and bulk uncovered (QL-R06/R02).
Disaster recovery	Partially aligned	DR runbooks, break-glass, restic restore drill done. Gap: no full cluster-loss rehearsal; single site.
Capacity planning	Partially aligned	Deterministic forecast in netframe_predict.py; GPU/quota dashboards. Gap: no formal capacity model.
Security review	Comparable	Multiple prior assessments (enterprise, reliability, security, red-team) with tracked remediation. Unusually mature.
Vulnerability management	Developing	Patch lag 63-142/node (QL-R13); no formal CVE tracking cadence. Improve: staged patch windows (QL-REC19).
Documentation	Partially aligned	Rich but drifting (Section 14).
Automation	Partially aligned	Ansible, systemd timers, CI, AI-ops. Gap: infra deploy stays manual by policy (defensible).
Testing	Partially aligned	netlab containerlab CI boots real routers and runs reachability/failover tests. Rare and impressive.
Release management	Developing	Config-as-code repos, but no versioned release process for infra.
Infrastructure as code	Partially aligned	Ansible + config-as-code monitoring; not fully declarative (no Terraform for PVE).
Service ownership	Developing	Single operator owns everything; clear but bus-factor 1.
SLIs / SLOs / error budgets	Material gap	Not defined. Improve: define a few SLOs (DNS availability, backup freshness) - high portfolio value.
Dependency mapping	Comparable	This review plus the dependency and recovery diagrams; netframe_monitor computes blast radius.
Business continuity	Developing	Break-glass + DR docs; single-site, single-operator constrain true BC.
Vendor lifecycle	Developing	EOL R730/EX3400/Mac-Mini tracked informally; no formal lifecycle register.
Auditability	Comparable	Hash-chained tamper-evident AI ledger, full git history, this audit trail.
Risk acceptance	Comparable	Risks explicitly accepted and documented (pinned kernels, raidz1, single-node monitoring).

16.1 What Makes This Unusually Strong for a Home Lab

NetFRAME runs three real, non-trivial production workloads (DUNE neutrino-physics research, a multi-tenant student GPU teaching cluster, and a self-hosted LLM inference platform) on properly segmented infrastructure. Defense-in-depth is not theoretical here: BMC isolation with an applied egress clamp, default-deny inter-VLAN policy, and least-privilege monitoring credentials are all live. The AI-operations system is safety-first by design (Section 10), with a deterministic evidence engine and human-gated remediation that would pass a serious enterprise review. Above all, the lab audits itself honestly (this review sits alongside four prior assessments and an in-flight reconciliation branch), which is the single strongest signal of operational maturity.

16.2 What Would Concern an Enterprise Hiring Manager

Bus factor of one: every design, credential, and recovery path depends on a single operator. There are no defined SLOs or error budgets, so reliability is asserted rather than measured. The environment is single-site with backups still short of 3-2-1 (QL-R03), and a meaningful set of security remediations remain open or blocked (registry auth, NetworkPolicies, guest SIEM coverage). Documentation drift means the public-facing docs would mislead a new hire until the reconciliation lands.

16.3 Portfolio Framing

For a candidate transitioning into network and infrastructure engineering, this environment demonstrates enterprise judgment far beyond the typical home lab: real segmentation, real DR rehearsal, real capacity and security review, and honest risk acceptance. The highest-leverage additions for recruiter and hiring-manager impact are defining two or three SLOs, closing the 3-2-1 gap, and landing the documentation reconciliation so the public repo reads as cleanly as the operator knows the estate to be.

17 Failure-Mode and Dependency Analysis

This section walks the credible failure scenarios for the estate. It is analytical only: **no failure injection was performed**, consistent with the read-only mandate. The dependency structure it draws on is in `diagrams/service-dependencies.mmd` and the recovery order in `diagrams/recovery-dependencies.mmd`.

The single most important structural observation is a small set of keystone dependencies: OPNsense (routing/DHCP/DNS egress), Randy (all backups, NFS, and k8s storage), the single Unbound resolver, and the co-located observability stack. Most high-impact scenarios below trace back to one of these four.

Scenario	Detection	User impact	Response / mitigation gap	Safe test?
Loss of 1 PVE node	InstanceDown, quorum votes	guests on that node down until manual action	Quorum holds (4/7+); no HA groups so restart guests elsewhere manually (QL-R18)	Yes (drain in window)
Loss of 2 PVE nodes	quorum + InstanceDown	possible quorum loss depending on nodes	5/7 still quorate; below 4 votes cluster read-only; recover a node	No
Loss of quorum	ClusterNotQuorate	config read-only, no guest changes	Restore a node or set expected votes; documented	No
Loss of OPNsense (QL-R04)	InstanceDown, WAN alerts, user reports	whole LAN: no routing/DHCP/egress	Serial console DR + age config restore; CARP HA planned (QL-REC10)	No
Loss of DNS	Pihole probes	name resolution fails for single-resolver clients	Dual Pi-hole HA + both handed out; single Unbound is the residual risk (R-03)	Yes (fail one Pi-hole)
Loss of DHCP	user reports, Kea health	new leases fail; static-in-container svcs unaffected	Kea on OPNsense; critical hosts static; restore OPNsense	Partial
Loss of EX3400	everything dark, no alert egress	full L2 outage	No switch redundancy; config not captured (QL-R16); manual rebuild	No
Loss of a trunk	link/port down (if monitored)	VLAN reachability degraded	Redundant UniFi path partly lost (xe-0/2/3 down, DR-14)	Yes (in window)
Loss of a corosync link	corosync ring status	ring degraded, quorum may wobble	knet single-link per node; ring on stable VLAN 1	No
Loss of Randy / storage (QL-R05)	InstanceDown, NFS/PBS errors	backups, NFS /data, k8s PVs all gone at once	Second target planned (R730xd); no live failover today	No
Loss of DS4246	zpool import fail, bulk offline	bulk data (research+media) unavailable	Single enclosure SPOF; bulk has no second copy (QL-R02)	No
Loss of Jarvis	NetframeMonitorStale, llm down	LLM inference + AI-ops reporting down	Deterministic alerter still runs; non-critical to core infra	Yes (stop svc)
Loss of Quarky-Lab	InstanceDown, SLURM down	research + student GPU + Wazuh down	Wazuh migratable; workspace pool local; students queue	No

Scenario	Detection	User impact	Response / mitigation gap	Safe test?
Loss of Grafana (QL-R10)	dead-man heartbeat	no metrics/logs/alert evaluation	Stack co-located on CT103; heartbeat can catch total loss	Yes (in window)
Loss of Headscale	tailnet coordination fails	admin mesh only; no user impact	Existing nodes keep working; rebuild from backup	Yes
Loss of NPM	reverse-proxy 502s	public hostnames down; internal IPs still work	Restore CT101; static backends reachable directly	Yes
Loss of internet	WAN alerts	no egress; alerting via Discord also blinded (QL-R09)	WAN2 5G failover; add 2nd OOB alert channel (QL-REC14)	Yes (unplug WAN1)
Loss of 1 UPS (QL-R17)	UPS status via NUT	that plane loses ride-through	UPS-A at 9.5 min is the tighter margin; rebalance (QL-REC08)	Yes (on battery test)
Extended power outage	UPS runtime alerts	staged shutdown then full stop	NUT graceful shutdown to audit (R-07); startup order documented	No (real outage)
Battery failure	UPS self-test/charge alerts	no ride-through on that UPS	Both 100% now; batteries age unknown; schedule test	No (needs approval)
Certificate expiry	RegistryCertExpiring (registry only)	TLS breaks for un-monitored certs	Broaden cert monitoring (QL-REC07); step-ca renewal	Yes
GitHub unavailable	push/pull fails	no config-as-code sync; infra unaffected	Local clones remain; manual deploy already the norm	Yes
Credential loss (QL-R08)	operator-detected	cannot unlock services	Break-glass age snapshot on Ares+Randy (offline)	Yes (test decrypt)
Accidental config deletion	drift/audit, user reports	service or rule removed	Git history + PBS + OPNsense config backup; restore	Yes (in test)
Failed k8s/PVE upgrade	node/pod failures	control plane or node degraded	Rolling upgrades keep quorum; kernel holds protect GPU nodes (QL-REC19)	Partial
Compromised student account	Wazuh/SLURM anomalies	contained by app-tainer + cgroups + no sudo	Batch-only, network=none, VRAM caps; guest SIEM blind spot (QL-R22)	Yes (test acct)
AI hallucination / unsafe rec	policy screen, audit ledger	none: no autonomous action possible	Deterministic evidence engine + human-gated allowlist remediation	Yes (eval harness)
Monitoring failure (QL-R11)	dead-man heartbeat	silent blind spot; a real risk here	ZfsPoolDegraded delivery unverified; verify path (QL-REC02)	Yes (synthetic)
Backup corruption	PBS verify failures	restore may fail	Weekly verify + GC; add stale/verify alerts (QL-REC07)	Yes (verify job)

Several of these are safely testable in production with a maintenance window (fail one Pi-hole, unplug WAN1 to exercise 5G failover, drain a node, restore a guest to a scratch VMID, test-decrypt break-glass). The genuinely unsafe tests - dual-node loss, storage loss, real power outage, battery failure - should only be modeled, never injected.

18 Drift and Contradiction Register

This phase compares five views of the estate: the state *declared* by the review specification, the state *configured* on devices, the state *observed* live on 2026-07-23, the state *documented* in the repository, and the *most likely correct* value adjudicated from the strongest evidence. No drift was resolved during this read-only review; each row records the recommended resolution and whether operator approval is required. The full field set is in `data/drift-register.csv`.

Two of the review specification's own "known corrections" are inverted versus live reality and must **not** be applied: DR-01 (GPU assignment) and DR-02 (OPNsense version direction). Live evidence and the repository agree against the specification on both.

ID	Item	Observed / configured	Documented	Correct	Appr.
DR-01	GPU assignment	QuarkyLab RTX 8000; Jarvis 2x RTX 6000 (E-02/E-03)	CLAUDE.md correct; 1 stale vault log	Live/repo	No
DR-02	OPNsense version	Not re-verified live (E-23)	25.1.12 vs 12 refs at 25.7	25.1.12	No
DR-03	Grafana node	CT103 on pve4 (E-06)	README/topology say pve3	pve4	No
DR-04	Headscale node	CT105 on pve5 (E-06)	README/ADR say pve3	pve5	No
DR-05	Randy CPU	E5-2690 v3 24c/48t	docs say v4 28c/56t	v3	No
DR-06	Randy Seagate drives	4 in raidz2-3 (E-06)	docs say 2 spare	4 in-pool	No
DR-07	bulk geometry	3x RAIDZ2 80 T (E-04)	2x8-wide 58.2 T	3-vdev 80 T	No
DR-08	EX3400 native-vlan-id	Not live (SSH broken E-22)	docs say unsupported	Supported (ELS)	No
DR-09	Homepage location	CT106 on pve3 (E-06)	pve1 CT104	pve3 CT106	No
DR-10	Randy PVE version	9.2.3 live (E-04)	docs say 9.1.1	9.2.3	No
DR-11	pve1 CT104 homepage	running (E-05)	docs say retired 06-24	stop/disable	Yes
DR-12	Ansible hardening	-- check only (E-24)	role exists	not enforced	Yes
DR-13	DS4246 cabling	Not live-inspected	repo vs spec differ	verify at HW	No
DR-14	xe-0/2/3 SFP link	DOWN (arch digest)	intended 10G	fiber SFP+ fix	Yes

18.1 Interpretation

DR-03 through DR-09 are the classic one-directional documentation drift: an unambiguous, doc-recorded canonical value exists and the fix is doc-only, already staged on the reconciliation branch. DR-01 and DR-02 are the reverse trap, where the incoming specification is stale and applying its "correction" would introduce error; live `nvidia-smi` (E-02/E-03) and the operator's own AAR settle both.

Four items are newly surfaced by this review's live pass rather than the prior repo audit. DR-10 (Randy running PVE 9.2.3 while every document still says 9.1.1) is a benign but real configuration-versus-documentation gap and should be reconciled after confirming the upgrade was intended. DR-11 (the retired homepage container CT104 still running on pve1) and DR-12 (the hardening role never actually applied to nodes, only -- check verified) are drift between assumed and actual runtime state and both require approval to resolve because they are service- or configuration-affecting. DR-14 (the UniFi SFP uplink down on an EEPROM speed mismatch) is a lost redundant path needing a hardware change.

19 Risk Register

Severity follows a fixed model and is deliberately not inflated. **CRITICAL** an immediate, credible risk of catastrophic outage, unrecoverable loss, or serious compromise. **HIGH** a significant risk requiring prompt corrective planning. **MEDIUM** a material weakness to schedule. **LOW** an improvement opportunity or limited risk. **INFO** a documentation or visibility observation. Every risk is tied to environment evidence; the full field set (description, existing controls, residual risk, validation method, confidence) is in `data/risk-register.csv`.

19.1 Summary

0 Critical, 7 High, 12 Medium, 3 Low. No finding meets the Critical bar today: the most dangerous condition (a faulted drive in the bulk pool) is still tolerated by RAIDZ2 with no data errors and a replacement in progress. It is rated High rather than Critical, but it is the top operational priority because QL-R01 pairs with QL-R02 (the bulk pool has no second copy) to form the single most serious combined exposure in the estate: one more drive failure in the same vdev before replacement would mean permanent loss of data that exists nowhere else.

19.2 High

ID	Title	Category	Treatment
QL-R01	Faulted bulk drive in a degraded RAIDZ2 vdev	Storage	Replace mpathv + scrub (QL-REC01)

ID	Title	Category	Treatment
QL-R02	bulk pool has no second copy	Backup	Add bulk/fernanda to PBS or B2 (QL-REC03)
QL-R03	All backups on one node/rack/UPS (3-2-1 gap)	Backup/DR	Second target R730xd or B2 (QL-REC03)
QL-R04	OPNsense is a single point of failure	Network	CARP HA pair (QL-REC10)
QL-R05	Randy is a single storage node	Storage	Second storage/backup target
QL-R11	ZfsPoolDegraded alert delivery unverified	Monitoring	Verify Discord delivery (QL-REC02)

QL-R01 through QL-R05 and QL-R11 are the High tier. QL-R11 is rated High because a silent critical alert is more dangerous than no alert: the degraded pool was in fact surfaced by a staleness alert, not by the pool-health rule, so the delivery path is unproven.

19.3 Medium

ID	Title	Category	Treatment
QL-R06	pve1 guests unprotected (no backup)	Backup	vzdump pve1 to Randy (QL-REC04)
QL-R07	GitHub PAT plaintext at rest on Ares	Security	Rotate + move to env (QL-REC05)
QL-R08	Secrets-recovery circular dependency	Security/DR	Keep break-glass current + test
QL-R09	Single WAN/Discord alert path	Monitoring	Add out-of-band channel (QL-REC14)
QL-R10	Observability stack co-located on one CT	Monitoring	External alert eval / heartbeat
QL-R12	Kernel/GPU hold asymmetry on Jarvis	Reliability	Mirror QuarkyLab holds (QL-REC12)
QL-R13	Fleet patch lag (63-142/node)	Security	Staged patch + reboots (QL-REC19)
QL-R14	Unauthenticated LAN AI/service endpoints	Security	Registry auth; keep locks (QL-REC15)
QL-R15	No NetworkPolicies / PSA in RKE2	Security	Default-deny + PSA (QL-REC15)
QL-R16	EX3400 config not captured; SSH broken	Network/DR	Reconcile SSH + Oxidized (QL-REC09)
QL-R17	UPS-A overloaded / short runtime	Power	Rebalance loads (QL-REC08)
QL-R22	Wazuh coverage limited to hypervisors	Security	Guest agents + detections

19.4 Low and Accepted

ID	Title	Category	Disposition
QL-R18	HA fencing in standby (no auto recovery)	Availability	Decide per guest; may be accepted
QL-R20	Randy PVE version drift (9.2.3 vs docs)	Configuration	Doc reconcile (Low)
QL-R21	raidz1 on aged disks	Storage	Accepted; monitor SMART, spares

Every entry above traces to a cited evidence item and, where verifiable, was confirmed by at least two independent read-only methods (for example QL-R01 by live `zpool status`, the prior audit capture, and the DS4246 slot-15 incident record). Confidence per risk is recorded in the CSV; the two items carrying Medium confidence are QL-R05 and QL-R11, both constrained by what a read-only pass can prove.

20 Recommendations

Every recommendation below is tied to specific evidence from this environment. A finding is not automatically a recommendation, and stable systems are not changed without a compelling reason. This review executed **none** of these actions; all await explicit operator approval. The full machine-readable set with rollback, validation, effort, and impact fields is in `data/recommendation-register.csv`.

20.1 By Class

ID	Class	Action	Approval
QL-REC02	Immediate follow-up	Verify ZfsPoolDegraded reached Discord	No (read)
QL-REC11	Immediate follow-up	Land the D-01..D-10 documentation PR	No (docs)
QL-REC01	Urgent corrective	Replace faulted bulk drive + scrub	Yes
QL-REC05	Urgent corrective	Rotate at-rest creds on Ares	Yes
QL-REC04	Next window	Back up pve1 guests to Randy	Yes
QL-REC08	Next window	Rebalance UPS loads + verify NUT shutdown	Yes

ID	Class	Action	Approval
QL-REC12	Next window	Mirror Jarvis GPU package holds	Yes
QL-REC19	Next window	Staged fleet patch + reboots	Yes
QL-REC03	Near-term	Second backup copy (R730xd or B2)	Yes
QL-REC06	Near-term	Deploy prometheus-pve-exporter	Yes
QL-REC07	Near-term	Add quorum/backup-stale/verify/cert rules	Yes
QL-REC09	Near-term	Reconcile EX3400 SSH + deploy Oxidized	Yes
QL-REC14	Near-term	Second out-of-band alert channel	No
QL-REC18	Near-term	.gitignore in 6 repos + OPSEC review	No
QL-REC10	Medium-term	Build OPNsense CARP HA pair	Yes
QL-REC13	Medium-term	Wire remediation gate to evidence engine	No
QL-REC15	Medium-term	Registry auth + RKE2 NetworkPolicies	No
QL-REC16	Optional	Mask failed openipmi.service on pve1	Yes
QL-REC17	Optional	Confirm + retire duplicate homepage CT104	Yes
QL-REC20	No action	Keep pinned GPU kernels (do not upgrade)	No
-	No action	Do not enforce VLAN-1 egress lockdown yet	No

20.2 Detailed Treatment of the Top Items

QL-REC01 - Replace the faulted bulk drive (Urgent)

Problem: mpathv is FAULTED in raidz2-2; that vdev now has one of two parity drives (QL-R01, E-04). **Action:** insert a ≥ 4 TB SAS drive into one of the two free DS4246 bays, then `zpool replace bulk mpathv <new-by-id>`, watch the resilver, and scrub after. **Prerequisite:** a spare drive; identify the bay for WWN 5000c500631a54fb. **Validation:** `zpool status` returns ONLINE and a clean scrub. **Rollback:** pull the new drive; the pool returns to its current degraded state, not worse. **Window:** none needed. **Confidence:** High.

QL-REC03 - Add a second backup copy (Near-term)

Problem: every backup lives only on Randy; bulk research data has no copy at all (QL-R02/R03). **Action:** PBS remote-sync to the planned R730xd second PBS, or `rclone/PBS` to Backblaze B2 for datastore plus bulk/fernanda. **Prerequisite:** the R730xd build or a B2 account. **Validation:** a test restore from the second copy. **Rollback:** remove the sync job. **Approval:** required. **Confidence:** High.

QL-REC05 - Rotate at-rest credentials on Ares (Urgent)

Problem: a live GitHub PAT sits in plaintext in `/github_helper.py`, plus `/apikey.txt` and `/.wazuh-api-newpw.txt` (QL-R07, E-17). None are committed, but all are usable if the workstation is compromised. **Action:** rotate the PAT, move it and the other secrets to an environment file or credential store. **Validation:** the old token is revoked and scripts read from the environment. **Rollback:** retain the old file until the new path is confirmed. **Approval:** required (touches GitHub/Wazuh). **Confidence:** High.

QL-REC10 - OPNsense CARP HA pair (Medium-term)

Problem: a single OPNsense VM on pve2 is a whole-network single point of failure (QL-R04). **Action:** follow the existing 2026-07-22 CARP plan: a second node, both on 25.7, a dedicated pfsync link, Kea DHCP HA, and a standalone backup host outside the cluster; the WAN-side CARP behavior is the hard open item to solve first. **Validation:** a failover test in a maintenance window. **Rollback:** documented in the plan. **Window:** required. **Confidence:** Medium.

20.3 Explicitly No Action

Keep the GPU-node kernels pinned at 6.14.11-9-pve; upgrading breaks the NVIDIA 550 stack (QL-REC20). Do not enforce the VLAN-1 egress lockdown yet: its prerequisites (UniFi discovery and multicast, RKE2 and container-registry outbound, CDN dependencies, administrative access paths, DHCP reservations, emergency access, current firewall API permissions, and a tested rollback) are not all satisfied, and this is a standing operator decision that this review respects.

20.4 Approval-Driven Next Steps

The recommended sequence, each pending operator sign-off, is: (1) the two immediate read-only follow-ups (QL-REC02 verify alert delivery, QL-REC11 land the docs PR); (2) the urgent corrective pair (QL-REC01 drive replacement, QL-REC05 credential rotation); (3) the next-window batch (QL-REC04 pve1 backup, QL-REC08 UPS rebalance, QL-REC12 Jarvis holds, QL-REC19 staged patching); then (4) the near- and medium-term engineering items led by QL-REC03 (second backup copy) and QL-REC10 (OPNsense HA). Approval for any one action does not authorize the others.

21 Limitations and Confidence

This was a read-only review conducted in a single working session. Its conclusions are only as strong as the read-only evidence that supports them, and several areas could not be examined to the depth a change-authorized or longer engagement would allow. The limitations below are stated plainly so that the reader can calibrate each conclusion.

21.1 Access and method limitations

- **Read-only ceiling.** No configuration-changing or service-affecting command was run, so any finding that would require mutating state to confirm (for example, an actual restore test, a failover, or a UPS self-test) is assessed from configuration and documentation rather than demonstrated behavior.
- **OPNsense not live-inspected (E-23).** By design, no OPNsense API credential was used and VM 100 was not touched. Firewall rules, the running version, DHCP reservations, and gateway status therefore rest on the repository and the operator's AARs. The version 25.1.12 is documented but not re-verified live; confidence in firewall specifics is **medium**.
- **EX3400 not live-inspected (E-22 / R-09).** SSH key authentication to the switch is broken (it fell through to password authentication and was denied), so switch config, VLAN membership, spanning-tree state, optics status, and error counters could not be pulled. Switch facts are **medium** confidence and rest on documentation; the one live-adjacent signal is the SNMP data the dashboard already collects.
- **Alert delivery not verifiable (QL-R11).** Whether the critical ZfsPoolDegraded rule actually reached Discord could not be confirmed, as the Grafana alert history requires authentication. This is called out as an open item (QL-REC02) rather than assumed either way.
- **Faulted-drive identity.** Live multipath -ll mpathv returned empty during this session (E-04); the drive's WWN (5000c500631a54fb) and model (Seagate ST4000NM0023) are taken from the prior audit, which is corroborated by the live FAULTED state and the NF-INC-2026-07-22 record. Confidence in the fault is **high**; in the exact device identity, **medium-high**.
- **Binary artifacts not text-scanned.** Roughly 50 committed PDFs (concentrated in the assessment repositories) were not decoded for embedded secrets. The identity/secrets and IPAM PDFs in the enterprise-assessment repo warrant a human eyes-on for sensitivity (E-17).
- **Point-in-time metrics.** Live CPU, memory, UPS load, and job-queue readings are instantaneous. UPS runtime in particular is a load-dependent estimate and will vary; the imbalance conclusion (QL-R17) is robust, but the exact minutes are a snapshot.
- **Prior-assessment age.** The enterprise (E-18) and reliability (E-20) assessments are roughly 9 to 10 days old; some of their items have since been remediated (for example the GPU-persistence power fix). Their findings are used as corroboration and history, reconciled against current state where possible.

21.2 Confidence by major conclusion

- **High:** cluster quorum and membership; GPU assignment; ZFS pool health including the bulk fault; backup freshness; UPS online state and the load imbalance; repository secrets-cleanliness of tracked content; the at-rest workstation credential findings.
- **Medium-high:** the faulted device's precise identity; storage geometry details on the DS4246; the AI-ops safety controls (read from code, not runtime-traced).
- **Medium:** network control-plane specifics (OPNsense and EX3400); alert-rule delivery; the true applied state of the Ansible hardening role, which is --check-only and not enforced on nodes (E-24 / DR-12).

No unauthorized change was made at any point during this review, and no discovered credential was used or reproduced in full.

A Command and Activity Log

Every action this review performed was read-only and low-risk. No command in the config-changing, service-affecting, destructive, or external-target categories was executed, and no data was modified on any system. Timestamps are 2026-07-23 (EDT/UTC mixed as captured). The full raw output is retained in the reviewer session, not in this repository, to avoid committing transient host detail.

Time	System	Command (summarized)	Purpose	Result
09:22	Ares (local)	git rev-parse, git submodule status, ls	Establish repo root, branch, submodules	master; Home-Lab on audit branch; 8 submodules
09:22	Ares (local)	date, hostname, uptime, os-release	Local host identity	ares, Debian 12, up 15h
09:23	Ares (local)	read .claude memory + MEMORY.md	Load operational memory	11 memories read
09:23	Ares (local)	read Home-Lab CLAUDE.md, audit/	Guidance + prior audit basis	audit commit 65f3681
09:24	11 hosts	ping -c1 sweep	Reachability	all 11 UP (incl. VLAN 30)
09:27	pve3	pvecm status, corosync-cfgtool -s	Quorum + corosync ring	7/7 quorate, all links connected
09:27	QuarkyLab	nvidia-smi, zpool, qm list, sinfo	GPU, ZFS, guests, SLURM	RTX 8000 46GB; workspace ONLINE; VM104; student+research partitions
09:27	Jarvis	nvidia-smi, zpool, systemctl is-active	GPU, ZFS, AI services	2x RTX 6000; tank+scratch ONLINE; llm_router+ollama active
09:27	Randy	zpool list, zpool status -x/bulk	ZFS health	bulk mpathv FAULTED; datastore ONLINE
09:27	pve1 (.193)	pveversion, pct list, apt list	Standalone guests + patch lag	PVE 9.1.9; CT103+CT104 running; 142 pending
09:28	pve3	pvesh /cluster/resources, storage.cfg, ha-manager, pvesr	Cluster guests, storage, HA, replication	14 guests; 2 PBS targets; fencing standby; no replication
09:28	Randy	proxmox-backup-manager, zpool status datastore, multipath -ll	PBS + datastore geometry + drive id	datastore 4x RAIDZ2; multipath -ll returned empty
09:28	EX3400	ssh ex3400 (show version)	Switch state	DENIED (publickey) - confirms R-09
09:30	pve3	upsc trippelite, upsc midatlantic	UPS state	UPS-A 59% / ~9.5 min; UPS-B 36% / ~29 min
09:30	Randy	ls -dt /datastore/..., df -h	Backup freshness	14 snapshots dated 2026-07-23; 97G / 23T

All approvals for change remain outstanding by design: this review proposed actions (Section 20) but executed none of them. The environment was left exactly as found.

B Asset Register

Physical and virtual-host inventory, live-confirmed 2026-07-23 (evidence E-01 through E-06). Full field set is in data/asset-register.csv.

ID	Host	Role	CPU / GPU	IP / VLAN	Status and notes
H-01	pve1	Standalone Pi-hole / homepage	Intel (Mac Mini, runs PVE) / none	.10.193 / v1	up; NOT a cluster member; PVE 9.1.9; 142 pending; CT103 primary Pi-hole + CT104; NO backup job
H-02	pve2	OPNsense host + node	i7-8700 / none	.10.204 / v1	up; hosts OPNsense VM100 (edge SPOF); reboot-required; excluded from hardening
H-03	pve3	Core services + rke2-cp1	i7-8700 / none	.10.201 / v1	up; NPM/Vaultwarden/Homepage/OpenWebUI + VM201; only trunked EliteDesk; e1000e TSO/GSO-off mitigation
H-04	pve4	Grafana stack + rke2-cp2	i5-7500T / none	.10.202 / v1	up; CT103 Grafana/Prom/Loki/InfluxDB + VM202; sustained load ~5

ID	Host	Role	CPU / GPU	IP / VLAN	Status and notes
H-05	pve5	Headscale, Pi-hole2, rke2-cp3, HAOS	i5-7500T / none	.10.203 / v1	up; CT105 + CT108 + VM110 + VM203
H-06	QuarkyLab	ML / DUNE / student GPU	2x E5-2699 v4 / RTX 8000 48GB	.10.179 (+.30.179) / v1+30	up; kernel+nvidia-* held (6.14.11-9-pve); workspace raidz1 10.9T; VM104 Wazuh; SLURM
H-07	Jarvis	LLM inference	2x E5-2687W v4 / 2x RTX 6000 24GB	.10.31 (+.30.31) / v1+30	up; kernel held (nvidia not held - narrower); tank 9.09T + scratch; llm_router+ollama
H-08	Randy	Storage / PBS / Jellyfin / k8s worker	2x E5-2690 v3 / RX 580	.10.187 (+.30.187) / v1+30	up; datastore 36.7T ONLINE; bulk 80T DEGRADED (mpathv); PBS; live PVE 9.2.3 vs docs 9.1.1 (DR-10)
H-09	DS4246	JBOD disk shelf	NetApp 24-bay / n/a	dual SAS to Randy	degraded; 22x 4TB (2 free bays); slot 15 dead 2026-07-21; single enclosure SPOF
N-01	EX3400-48P	Core switch	JunOS 23.4R2-S7.4	.10.50 / all	up; RSTP root; dual PSU; SSH key auth broken (R-09); xe-0/2/3 DOWN (SFP mismatch)
N-02	UniFi USW-24	Access switch	UniFi	trunk ge-0/0/46	up
N-03	UniFi UDR	Wireless controller / AP	UniFi	.10.2 / v1	up; NOT the WAN edge (common misread)
N-04	EX2300-48P	Lab access switch	JunOS	trunk ge-0/0/45	up; lab / CCNA
N-05	APC AP7901	Managed PDU	APC	ge-0/0/38 / v1	up
P-01	UPS-B	Middle Atlantic OL2200R	feeds R730s+Randy+DS4246	n/a	up; OL, 100%, load 36%, ~29 min
P-02	UPS-A	Tripp Lite SMART1500	feeds switch+UniFi+EliteDesks	n/a	up; OL, 100%, load 59%, ~9.5 min (imbalanced)
W-01	Ares	Admin workstation	Intel laptop / Debian 12	.10.199 (+.20.199) / v1+20	up; ansible/backup/dashboard control node; timers; hourly .claude backup

C Service Catalog

Service inventory with placement, exposure, and backup posture. Full field set (auth, TLS, monitoring source, recovery doc) is in data/service-catalog.csv.

ID	Service	Host / guest	IP : port	External	Backup / state
S-01	OPNsense	pve2 / VM100	.10.1 : 443/53	LAN edge (dual-WAN)	randy-pbs 03:00 / running
S-02	Pi-hole primary	pve1 / CT103	.10.177 : 53/80	no	NONE (pve1 unbacked) / running
S-03	Pi-hole secondary	pve5 / CT108	.10.178 : 53/80	no	randy-pbs / running
S-04	Unbound	pve2 / VM100	.10.1 : 53	no	with OPNsense / running
S-05	Headscale	pve5 / CT105	.10.186 : 8080	internal	randy-pbs / running
S-06	Nginx Proxy Manager	pve3 / CT101	.10.181 : 80/443/81	yes (proxy front)	randy-pbs / running
S-07	Vaultwarden	pve3 / CT102	.30.182 : 80	yes (vault.kylemason.org)	randy-pbs / running
S-08	Homepage	pve3 / CT106	.10.148 : 3000	yes (homepage.kylemason.org)	randy-pbs / running
S-09	Homepage (dead)	pve1 / CT104	-	no	NONE / running (should be off, R-12)
S-10	Open WebUI	pve3 / CT107	.30.185 : 8080	yes (chat.netframe.local)	randy-pbs / running
S-11	Grafana stack	pve4 / CT103	.10.183 : 3000	yes (grafana.kylemason.org)	randy-pbs / running
S-12	Prometheus	pve4 / CT103	.10.183 : 9090	no	with Grafana / running
S-13	Loki	pve4 / CT103	.10.183 : 3100	no	with Grafana / running (auth_enabled=false)
S-14	Scrutiny	pve4 / CT103	.10.183 : 8080	no	with Grafana / running
S-15	Wazuh	QuarkyLab / VM104	.10.184 : 443/9200/55000	yes (wazuh.kylemason.org)	randy-pbs-10g / running
S-16	step-ca	pve2	.10.204 : 443	no	with pve2 / active
S-17	PBS	Randy	.10.187 : 8007	no	GC+verify weekly / active (single target)

ID	Service	Host / guest	IP : port	External	Backup / state
S-18	Jellyfin	Randy	.10.187 : 8096	no	- / active
S-19	Ollama	Jarvis	.10.31 : 11434	LAN (unauth, AI-01)	GPU exporter / active
S-20	llm_router	Jarvis	.10.31 : 8000	yes (llm.netframe.local)	scripts in git / active (iptables-locked)
S-21	netframe-monitor	Jarvis	.10.31 : 8808/8809	internal	scripts in git / active (15 min)
S-22	Home Assistant	pve5 / VM110	.10.60 : 8123	internal	randy-pbs 03:30 / running
S-23	RKE2	pve3/4/5 / VM201-203	.10.51-53 (VIP .54) : 6443/9345	no	randy-pbs 04:00 / running
S-24	Private registry	RKE2	registry.netframe.local : 443	no	in-cluster / running (anon, K8S-01)
S-25	Cloudflare Tunnel	QuarkyLab	quarkylab.kylemason.org	yes (no inbound ports)	- / active
S-26	SLURM	QuarkyLab	.10.179	no	- / active (shard+MPS+apptainer)

D Port and VLAN Register

VLAN plan and EX3400 core-switch port assignment. The switch was not live-inspected this review (SSH key auth is broken, evidence E-22 / risk QL-R16); the port map below is reconstructed from repository topology docs and the 2026-06-25 VLAN-activation runbook, and should be re-verified once switch access is restored (QL-REC09).

D.1 VLANs

ID	Name	Subnet	Purpose
1	mgmt / default	192.168.10.0/24	Proxmox hosts, corosync, services, switch management (dense)
20	trusted / OOB	192.168.20.0/24	iDRAC x2 + IPMI (BMCs) + Ares jump leg; inbound clamped to Ares; egress-clamped
30	servers	192.168.30.0/24	NFS / PBS / GPU egress; dual-homed GPU and storage nodes; blocked to VLAN 1 except :9100
40	IoT	192.168.40.0/24	smart-home, IMUs; isolated, DNS pinhole only
50	VoIP	192.168.50.0/24	CP-8841 phones, FreePBX (deferred)
60	guest	192.168.60.0/24	isolated guest WiFi
70	lab	192.168.70.0/24	CCNA / experimental; standalone sandbox node

D.2 EX3400-48P port map

Port	Attachment	Mode	VLANs
ge-0/0/22	Jarvis nic2 (onboard 1G)	Trunk	native 1 (+ tagged 30, now vestigial)
ge-0/0/24	QuarkyLab nic2	Trunk	native 1 + tagged 30
ge-0/0/30, /32, /44	iDRAC / IPMI OOB	Access	20 only
ge-0/0/38	APC AP7901 PDU	Access	1
ge-0/0/41	Ares mgmt / jump	Dual	native 1 + tagged 20
ge-0/0/45	EX2300 uplink	Trunk	all
ge-0/0/46	UniFi USW-24 Port 24	Trunk	native 1 + tagged 20/30/40/50/60/70
xe-0/2/0	Randy nic3 (10G ConnectX-3)	Trunk	1 + 30 (storage / NFS)
xe-0/2/2	Jarvis enp132s0 (10G ConnectX)	Access	30 (dedicated storage / egress)
xe-0/2/3	UniFi SFP2 DAC	DOWN	EEPROM 10G/1G speed mismatch (DR-14); fix = fiber SFP+ both ends

Note: native-vlan-id is set at the physical-interface level on this ELS switch, and it IS supported (contrary to a stale note in several docs, D-08). RSTP root is the EX3400 (bridge priority 4096).

E Backup Matrix

Per-asset backup posture. Last-success dates are live-confirmed for 2026-07-23 (14 snapshots present, evidence E-08). Full field set (encryption, RTO/RPO, failure domain) is in data/backup-matrix.csv. The two rows flagged **HIGH** are the material gaps.

Asset	Target	Schedule	Retention	Last ok	Gap
OPNsense VM100	randy-pbs + config repo	03:00 + 03:17	7d/4w	07-23	VM image only on Randy; config also age-encrypted to GitHub
CT101 NPM	randy-pbs	02:00	7d/4w	07-23	single copy
CT102 Vaultwarden	randy-pbs	02:00	7d/4w	07-23	secrets single-copy; break-glass offline mitigates
CT103 Grafana	randy-pbs	02:00	7d/4w	07-23	single copy
CT105 Headscale	randy-pbs	02:00	7d/4w	07-23	single copy
CT106 Homepage	randy-pbs	02:00	7d/4w	07-23	single copy
CT107 OpenWebUI	randy-pbs	02:00	7d/4w	07-23	single copy
CT108 pihole2	randy-pbs	02:00	7d/4w	07-23	mirrors CT103 config
VM104 Wazuh	randy-pbs-10g	03:00	7d/4w	07-23	single copy
VM110 HomeAssistant	randy-pbs	03:30	7d/4w	07-23	single copy
VM201/202/203	randy-pbs	04:00	7d/4w	07-23	etcd not separately backed up
RKE2					
QuarkyLab workspace	PBS host backup	05:30	per-job	07-23	fernanda refreservation floor
CT103/CT104 pve1	NONE	none	none	NEVER	HIGH NO BACKUP; config rebuildable from CT108 (R-03)
bulk pool	NONE	n/a	n/a	NEVER	HIGH NO SECOND COPY of media + fernanda research (R-04)
Ares .claude	randy:/datastore	hourly	accumulate	07-23	secrets excluded
Ares /home	restic to randy	04:03	restic	07-23	lands ON the degraded bulk pool

All production backups currently reside only on Randy (one PBS node, one rack, one UPS bus): effectively 1-1-1, not 3-2-1 (QL-R03). The Ares workstation restic target sits on the degraded bulk pool, compounding QL-R01. A second, independent copy is the single highest-value recoverability improvement (QL-REC03).

F Alert Matrix

Grafana Unified Alerting rules (no Alertmanager; all egress to Discord). Full field set (evidence basis, expected response, runbook, false-positive/negative risk) is in `data/alert-matrix.csv`. The `GAP-*` rows are alerts that should exist but do not.

Alert	Source	Sev	Threshold	Channel	Disposition
ClusterNotQuorate	corosync textfile	crit	quorate <1, 2m	discord-alerts	Keep
ClusterVotesBelowExpected	corosync textfile	warn	missing votes >0, 5m	discord-alerts	Keep
InstanceDown	node / blackbox	crit	up <1, 2m	discord-alerts	Keep
PiholePrimaryDown	blackbox DNS	crit	probe .177 <1, 2m	discord-alerts	Keep
PiholeSecondaryDown	blackbox DNS	warn	probe .178 <1, 2m	discord-alerts	Keep
ZfsPoolDegraded	zpool textfile	crit	pool_health >0, 5m	discord-alerts	Verify delivery (R-02)
GpuTempHigh	nvidia-smi	warn	temp >87C, 5m	discord-alerts	Keep
GpuMemoryHigh	nvidia-smi	warn	VRAM % >90, 10m	discord-alerts	Keep
DiskAlmostFull	node_filesystem	warn	root % >90, 10m	discord-alerts	Extend to pools
LowMemory	node_memory	warn	MemAvail % <8, 10m	discord-alerts	Keep
RegistryCertExpiringSoon	blackbox tls	warn	cert <12h, 15m	discord-alerts	Broaden to all certs
BackupVerifyReportStale	Randy textfile	warn	age >26h, 15m	discord-alerts	Keep (caught mpathv)
BackupVerifyFailing	Randy textfile	warn	overall_pass <1, 5m	discord-alerts	Consider critical
AresBackupStale	Randy textfile	warn	age >26h, 15m	discord-alerts	Keep
NetframeMonitorStale	Jarvis heartbeat	warn	age >1h, dead-man	discord-alerts	Keep
UPS battery low	PeaNUT	crit	charge <50, 2m	discord-ups	Keep
UPS runtime low	PeaNUT	crit	runtime <300s, 1m	discord-ups	Keep
PrimaryWANDown	pve2 textfile	crit	WAN_GW <1, 1m	discord-alerts	Keep
FirewallBlockRateSpike	Loki filterlog	warn	>500 blocks/5m	discord-alerts	Keep
NetworkDeviceConfigChange	loki syslog	warn	commit >0	discord-alerts	Keep
NetworkSyslogSilent	Loki stream	warn	<50 lines/15m, dead-man	discord-alerts	Keep
GAP-quorum-guest	(none)	-	per-guest down (needs pve-exporter)	-	Missing (REC06/07)

Alert	Source	Sev	Threshold	Channel	Disposition
GAP-pbs-verify-stale	(none)	-	PBS verify job failed / stale	-	Missing (REC07)
GAP-host-auth-failure	(none)	-	repeated auth failures per host	-	Missing (DETECT-01c)
GAP-cert-expiry-broad	(none)	-	any netframe.local / LE cert near expiry	-	Missing (only registry today)
GAP-2nd-channel	(none)	-	alert egress independent of WAN/Discord	-	Missing (REC14)

G Dependency Register

Critical-service dependencies and the impact of losing each dependency. Derived from the service catalog (Appendix C) and diagrams/service-dependencies.mmd.

Service	Depends on	Impact if dependency lost	Mitigation
All LAN clients	OPNsense (VM100, pve2)	No routing, DHCP, or inter-VLAN; whole network down	Serial-console DR; nightly age config backup; CARP HA planned (QL-R04)
All name resolution	Pi-hole .177/.178 to Unbound on .1	Clients lose DNS; single Unbound is the recursion choke point	Dual Pi-hole HA (DHCP hands out both); Unbound still single (enterprise R-03)
All backups, NFS, k8s PVs	Randy (storage / PBS)	Simultaneous loss of backups, research NFS, and k8s storage	ZFS raidz2; UPS; second target planned (QL-R05 / QL-REC03)
Cluster control plane	corosync quorum (4/7)	Loss of quorum freezes cluster config and HA	7 votes, healthy; corosync on stable VLAN 1 L2 (E-01)
Everything	UPS-A and UPS-B	Unclean shutdown; data-at-risk on abrupt power loss	NUT monitored; verify graceful-shutdown coverage (QL-R17 / R-07)
Full secret set	Vaultwarden (CT102, in-cluster)	Cluster-loss recovery cannot read the secrets it needs	Break-glass age snapshot on Ares + Randy, offline (QL-R08)
Metrics, logs, alerting	Grafana stack (CT103, pve4)	Blind to all metrics/logs and no alert evaluation	Dead-man heartbeat can catch total loss; single-CT SPOF (QL-R10)
Alert delivery	WAN + Discord webhook	No alerts reach the operator during a WAN/Discord outage	Dead-man alerts; add 2nd channel (QL-R09 / QL-REC14)
LLM chat and RAG	Ollama on Jarvis GPUs	Chat/RAG down; local inference unavailable	llm_router auto-fallback to Claude API; deterministic monitoring unaffected
Research and student jobs	SLURM + Randy NFS + GPU	No new jobs; running jobs may fail on NFS loss	Local base.sif copy; refreservation floor; preemption (QL-R02 data single-copy)
Kubernetes workloads	RKE2 etcd + Randy NFS SC	Control plane or PV loss stops workloads	3-node etcd quorum; VM-level backups; no etcd-native backup (gap)
Internal TLS (netframe.local)	step-ca (pve2)	Certs cannot renew; registry / internal TLS breaks	RegistryCertExpiringSoon alert; CA password in Vaultwarden